# GLOBAL ACADEMY OF TECHNOLOGY

**(An Autonomous Institution Approved by UGC/AICTE/GOK, Affiliated to VTU, Belagavi, Karnataka)**

**Accredited by NAAC with 'A' grade**

**RR Nagar, Bengaluru – 560 098**

**Department of Computer Science & Engineering (AI&ML)**

# COMPUTER NETWORKS

# Laboratory Manual

**V Semester**

**Course Code: BCI24502**

**Academic Year 2026-27**

**Version 2.0**

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# Global Academy of Technology, Bengaluru - 98

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**Department of Computer Science & Engineering (AI&ML)**

# COMPUTER NETWORKS

# (BCI24502)

## Fifth Semester, B.E

***Name:*** _____ ***USN:*** _____

***Section:*** _____ ***Branch:*** _____

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## Department of Computer Science & Engineering (AI&ML)

**This is to certify that Mr. / Ms._____**

**bearing USN _____of the department has satisfactorily completed the Experiments in_____ prescribed by the Global Academy of Technology (An Autonomous Institute under VTU), in the laboratory of this college in the year _____.**

| | MAXMARKS | SCORED MARKS |
|---|---|---|
| CIE | | |
| IA | | |
| FINAL | | |

**Signature of faculty in charge** **Signature of HoD**

# DOCUMENT LOG

| | |
|---|---|
| **Name of the document** | **Computer Networks** |
| **Scheme** | **2024** |
| **Current version number** | **Version 2.0** |
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| **Prepared by** | **Dr. Shruthi B V, Ms. Larisha Sandra, Dept. of CSE(AI&ML)** |
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# Vision of the Institute

Become a premier institution imparting quality education in engineering and management to meet the changing needs of society.

# Mission of the Institute

M1: Create environment conductive for continuous learning through quality teaching and learning processes supported by modern infrastructure.

M2: Promote research and innovation through collaboration with industries.

M3: Inculcate ethical values and environmental conscious through holistic education programs.

## Vision of the Department

To Become a leading hub of excellence in education, research in the field of Computer Science and Engineering providing AI-driven solutions with holistic development for the needs of the Society.

## Mission of the Department

- To Provide aspiring engineers for industry and academia by offering excellent education in emerging AI techniques.
- To equip value-added technical and research-oriented education by satisfying societal needs
- To educate with professional integrity values and ethics for the environment awareness.

## PROGRAM EDUCATIONAL OBJECTIVES(PEOs)

- **PEO1:** Design and Develop innovative intelligent systems for the welfare of the Society.
- **PEO2:** Engage in lifelong learning process through higher education and to inculcate innovative ideas in research field.
- **PEO3:** Lead the IT Industry with management and Entrepreneurship skills.

## PROGRAM SPECIFIC OUTCOMES(PSOs)

- **PSO1:** To Produce graduates with industry-ready skills with the knowledge of Computer Science & Machine Learning technology based to solve real world problems.
- **PSO2:** Ability to develop many successful applications and designing efficient algorithms for intelligent systems within the realm of artificial intelligence incorporating in data analytics, Natural language processing and Internet of Things.

## PROGRAM OUTCOMES (PO's)

**PO1: Engineering Knowledge:**
Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

**PO2: Problem Analysis:**
Identify, formulate, review research literature and analyse complex engineering problems reaching substantiated conclusions with consideration for sustainable development (WK1 to WK4).

**PO3: Design/Development of Solutions:**
Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required (WK5).

**PO4: Conduct Investigations of Complex Problems:**
Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions (WK8).

**PO5: Engineering Tool Usage:**
Create, select, and apply appropriate techniques, resources, and modern engineering & IT tools, including prediction, and modelling recognizing their limitations to solve complex engineering problems (WK2 and WK6).

**PO6: The Engineer and The World:**
Analyse and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment (WK1, WK5, and WK7).

**PO7: Ethics:**
Apply ethical principles and commit to professional ethics, human values, diversity, and inclusion; adhere to national & international laws (WK9).

**PO8: Individual and Collaborative Teamwork:**
Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

**PO9: Communication:**
Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

**PO10: Project Management and Finance:**
Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

**PO11: Life-Long Learning:**
Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies, and iii) critical thinking in the broadest context of technological change (WK8).

# COURSE OUTCOMES:

**Upon successful completion of this course, students are able to:**

| | |
|---|---|
| **CO1** | Configure and verify basic network devices, IP addressing, and connectivity using Cisco Packet Tracer. |
| **CO2** | Implement and analyze static and dynamic routing protocols such as Static Routing, RIP, and EIGRP in interconnected networks. |
| **CO3** | Configure and troubleshoot VLANs, Inter-VLAN Routing, VTP, and switch security mechanisms using network simulation tools. |
| **CO4** | Implement network security mechanisms including Standard and Extended ACLs, port security, and basic router and switch security configurations. |
| **CO5** | Analyze and troubleshoot network communication using simulation and packet-analysis tools such as Cisco Packet Tracer and Wireshark. |

# Syllabus

| | | |
|---|---|---|
| Department**: Computer Science and Engineering (AI & ML)** | | |
| Semester: **V** | Course Code: **BCI24502** | Contact Hrs /week: **2** |
| Course Description: **Computer Networks** | | No. of Credits:**4**<br>L: T: P: S = **3:0:2:0** |
| Course Category: **IPCC** | | Total no. of Hours = 50 |
| CIE: **50 Marks** | SEE: **50 Marks** | Exam Hours: **03** |
| Course Pre-requisites: Basic knowledge of Computer Networks, IP addressing, and networking fundamentals. | | |

**Course Learning Objectives:**

The course will enable students to:

| | |
|---|---|
| CLO1 | Understand the fundamentals of computer networks, IP addressing, network devices, and basic network configuration using Cisco Packet Tracer. |
| CLO2 | Configure and analyze static and dynamic routing techniques to establish communication between interconnected networks. |
| CLO3 | Configure and troubleshoot VLANs, Inter-VLAN Routing, VTP, and basic switching operations using Cisco Packet Tracer. |
| CLO4 | Understand the basic processing unit and memory system of a computer Implement basic network security mechanisms such as switch port security, router and switch access control, and Standard and Extended ACLs. |
| CLO5 | Analyze network communication and troubleshoot network traffic using Cisco Packet Tracer and Wireshark, including ARP and packet-flow analysis. |

| Sl. No. | EXPERIMENTS |
|---|---|
| 1 | Configure a switch with basic security mechanisms using Cisco Packet Tracer. |
| 2 | Configure a router using basic IOS commands and verify network connectivity. |
| 3 | Configure IPv4 addresses and implement default routing between networks. |
| 4 | Implement static routing between multiple networks using Cisco Packet Tracer. |
| 5 | Implement dynamic routing using RIP and verify the routing table. |
| 6 | Implement EIGRP routing protocol and verify network connectivity. |
| 7 | Configure VLANs and implement Inter-VLAN routing using Cisco Packet Tracer. |
| 8 | Implement standard Access Control Lists (ACLs) to control network access. |
| 9 | Implement extended ACLs and analyze network traffic using Wireshark. |
| 10 | Capture and analyze ARP packets using Wireshark. |
| 11 | Configure advanced switch security mechanisms and verify the configuration. |
| 12 | Configure and troubleshoot router connectivity using Cisco Packet Tracer. |
| 13 | Switch Configuration and Security Implementation using Cisco Packet Tracer |
| 14 | Router Configuration using Basic Commands in Cisco Packet Tracer. |

**Mapping of CO-PO:**

| CO/PO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **CO1** | 3 | 3 | 3 | | 3 | | | 3 | | | | | 3 | |
| **CO2** | 3 | 3 | 3 | 3 | 3 | | | 3 | | | | | 3 | |
| **CO3** | 3 | 3 | 3 | 3 | 3 | | | 3 | | | | | 3 | |
| **CO4** | 3 | 3 | 3 | 3 | 3 | | | 3 | | | | | 3 | |
| **CO5** | 3 | 3 | 3 | 3 | 3 | | | 3 | | | | | 3 | |

**Low-1: Medium-2: High-3**

## MARKS DISTRIBUTION

| | Component | Marks | Total Marks |
|---|---|---|---|
| **CIE** | CIE Test-1 | **30** | **50** |
| | CIE Test-2 | **30** | |
| **Grand Total** | | | **100** |

## LAB EVALUATION PROCESS

**Continuous Internal Evaluation (CIE)**

| **Evaluation of CIE** | | |
|---|---|---|
| **S. No** | **Activity** | **Marks** |
| 1 | Average of Weekly Entries | 20 |
| 2 | Internal Assessment reduced to | 15 |
| **TOTAL** | | **15** |

# LAB RUBRICS

### Rubrics for Evaluation of Observation

| Attribute | Max. Marks | Good | Satisfactory | Poor |
|---|---|---|---|---|
| | | 4-5 | 2-3 | 0-1 |
| Writeup | 05 | Writes the program without errors. | Writes the program with few mistakes. | Unable to write the program. |
| | Max. Marks | 9-10 | 5-8 | 0-4 |
| Execution of program | 10 | • Debugs the program independently.<br>• Executed the program for all possible inputs. | • Works with little help from faculty.<br>• All possible input cases not covered. | • Unable to complete the execution of the program within the lab session. |
| | Max. Marks | 3-5 | 1-2 | 0 |
| Viva Voce | 05 | • Able to explain the logic of the program.<br>• Answered all questions. | • Partially understood the logic of the program.<br>• Answered few questions. | • Not understood the logic of the program.<br>• Not answering any questions |

### Rubrics for Evaluation of Record

| Attribute | Max. Marks | 9-10 | 5-8 | 0-4 |
|---|---|---|---|---|
| Complete Program with Algorithm (wherever applicable) and all possible Outputs | 10 | • Complete program, algorithm and outputs. | • Writes program with few mistakes<br>• Outputs are not clear.<br>• Late submissions. | • Not submitted the records.<br>• Incomplete program<br>• Not written algorithm |

### Rubrics for Evaluation of Internal Test

| Attribute | Max Marks | Good | Satisfactory | Poor |
|---|---|---|---|---|
| | | 3-5 | 1-2 | 0 |
| Writeup | 05 | • Completes source code.<br>• No syntax and logical errors.<br>• All possible inputs listed with expected output. | • Completes source code.<br>• Syntax and logical errors exist.<br>• All possible inputs listed with expected output. | • Incomplete source code.<br>• Change of program. |
| | Max Marks | 9-10 | 5-8 | 0-4 |
| Execution | 10 | Able to debug the program and get the right set of outputs. | • Program, executed for only few input cases.<br>• With few errors. | • Program with many errors.<br>• Not executed. |
| | Max Marks | 3-5 | 1-2 | 0 |
| Viva Voce | 05 | Answering all questions. | Answering few questions. | Not Able to answer basic questions. |

# Evaluation Sheet

| Sl. No | Date | Particulars | Page No. | Marks | Total (20M) | Sign |
|---|---|---|---|---|---|---|
| | | | | W+E+V (5M+10M+5M) | | |
| 1 | | | | | | |
| 2 | | | | | | |
| 3 | | | | | | |
| 4 | | | | | | |
| 5 | | | | | | |
| 6 | | | | | | |
| 7 | | | | | | |
| 8 | | | | | | |
| 9 | | | | | | |
| 10 | | | | | | |
| Average (Max. 20) | | | | | | |

**Signature - Faculty In-charge**

# INTRODUCTION

Computer Networks is a fundamental area of computer science that focuses on the communication and interconnection of computers, networking devices, and systems. This lab provides a hands-on platform for understanding network configuration, communication, routing, switching, security, and network traffic analysis using simulation and packet-analysis tools.

The lab uses **Cisco Packet Tracer** to provide a simulated environment for configuring routers, switches, PCs, and other network devices. Students learn how IP addressing, routing protocols, VLANs, and network security mechanisms operate in practical network environments.

The experiments covered in this laboratory include:

**1. IP Addressing and Basic Network Configuration**

IP addressing enables devices to communicate within and across networks. Students configure IPv4 addresses, subnet masks, default gateways, and router interfaces.

**Application:** Fundamental for establishing communication between computers and network devices.

**2. Router Configuration and Basic Routing**

Routers connect different networks and forward packets based on destination IP addresses. Basic router configuration includes interface addressing, hostname configuration, passwords, and connectivity verification.

**Application:** Used to interconnect departmental, organizational, and external networks.

**3. Static and Default Routing**

Static routing allows administrators to manually define routes between networks, while default routing provides a path for traffic destined for unknown networks.

**Application:** Suitable for small, stable networks and communication with external networks.

**4. Dynamic Routing**

Dynamic routing protocols allow routers to automatically exchange routing information and select appropriate paths.

**Application:** Used in growing and interconnected networks where manual route configuration becomes complex.

**5. RIP and EIGRP**

RIP and EIGRP are dynamic routing protocols used to exchange network information between routers. Students configure these protocols and verify learned routes and connectivity.

**Application:** Routing management in interconnected enterprise networks.

**6. VLAN and Inter-VLAN Routing**

VLANs logically divide a physical network into separate broadcast domains. Inter-VLAN routing enables communication between different VLANs using router or Layer 3 switching techniques.

**Application:** Departmental network segmentation for improved security, management, and performance.

**7. VTP and Switch Troubleshooting**

VTP helps manage VLAN information across switches, while troubleshooting techniques are used to identify VLAN, trunk, and connectivity problems.

**Application:** Centralized VLAN management and maintenance of switched networks.

**8. Access Control Lists**

Standard and Extended ACLs control network traffic according to defined security rules. Standard ACLs primarily use source addresses, while Extended ACLs can consider source, destination, protocol, and port information.

**Application:** Network traffic filtering and access control.

**9. Switch and Port Security**

Switch security mechanisms such as passwords, management interfaces, and port security help prevent unauthorized access to network resources.

**Application:** Protection of campus and enterprise switching environments.

**10. ARP and Network Address Resolution**

ARP resolves an IPv4 address into a corresponding MAC address so that devices can communicate at the data-link layer within a local network.

**Application:** Essential for communication between devices on Ethernet-based networks.

**11. Wireshark and Network Traffic Analysis**

Wireshark is used to capture and analyze network packets and examine protocols, addresses, ports, and communication behaviour.

**Application:** Network monitoring, troubleshooting, protocol analysis, and security investigation.

This version is aligned with the networking experiments you have prepared from Programs 1–14

## Tool Used: Cisco Packet Tracer & Wireshark

What is Cisco Packet Tracer?

Cisco Packet Tracer is a network simulation and visualization tool used to design, configure, and troubleshoot computer networks. It allows students to create virtual network topologies using routers, switches, PCs, servers, and other networking devices without requiring physical hardware.

**Key Features of Cisco Packet Tracer**

Network Topology Design: Users can create and connect routers, switches, PCs, servers, and other network devices in a logical workspace.

Device Configuration: Routers and switches can be configured using Cisco IOS commands and graphical configuration options.

Simulation Mode: Packets can be observed as they travel through the network. This helps students understand packet flow, routing, switching, and protocol operations.

Protocol Analysis: Students can observe protocols such as ICMP, ARP, RIP, and other network activities during simulation.

Troubleshooting: Packet Tracer can be used to identify configuration errors in IP addressing, routing, VLANs, trunks, ACLs, and network connectivity.

Security Configuration: The tool supports practical implementation of passwords, port security, VLANs, ACLs, and other basic network-security mechanisms.

What is Wireshark?

Wireshark is a network protocol analyzer used to capture and examine network packets. It allows users to inspect network communication at the packet level.

**Key Features of Wireshark**

Packet Capture: Captures network traffic from an available network interface.

Protocol Analysis: Allows examination of protocols such as ARP, ICMP, TCP, UDP, and HTTP.

Packet Filtering: Display filters can be used to locate specific types of network traffic, such as arp, icmp, or tcp.

Traffic Analysis: Users can examine source and destination IP addresses, MAC addresses, protocols, port numbers, and other packet information.

Network Troubleshooting: Wireshark helps identify communication problems and analyze network behaviour.

# LAB PROGRAMS

1. **Switch Configuration and Security Implementation using Cisco Packet**
   **A college campus network requires secure access to its internal systems. Unauthorized devices connecting to switches can compromise security. To prevent this, the network administrator configures a switch with basic settings, enables port security, and sets up passwords for secure access using Cisco Packet Tracer.**

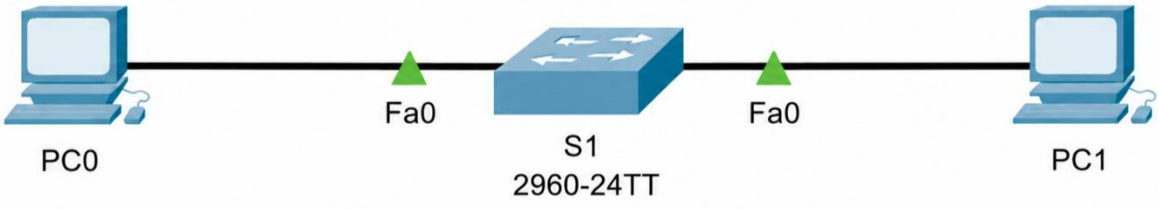
   **Aim**: - To configure a Cisco switch with basic security settings using Cisco Packet Tracer and verify the configuration.

   **Apparatus Required**: -

   1. Cisco Packet Tracer
   2. Cisco Switch
   3. PCs
   4. Copper Straight-Through Cables

A switch is a Data Link Layer device used to connect multiple devices within a Local Area Network (LAN). It forwards frames based on MAC addresses and helps reduce unnecessary network traffic. Basic switch configuration includes assigning a hostname, securing privileged access, configuring console and VTY access, encrypting passwords, and displaying a login warning message.

**Network Topology**




| Device | Interface | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|---|
| PC0 | FastEthernet0 | 192.168.1.10 | 255.255.255.0 | 192.168.1.253 |
| PC1 | FastEthernet0 | 192.168.1.20 | 255.255.255.0 | 192.168.1.253 |
| Switch S1 | VLAN 1 | 192.168.1.253 | 255.255.255.0 | - |

**Procedure**

1. Open Cisco Packet Tracer and place one Cisco 2960 switch and two PCs in the workspace.

2. Connect PC0 and PC1 to Switch S1 using Copper Straight-Through cables.
3. Configure the IP address of PC0:
   - IP Address: 192.168.1.10
   - Subnet Mask: 255.255.255.0
   - Default Gateway: 192.168.1.253
4. Configure the IP address of PC1:
   - IP Address: 192.168.1.20
   - Subnet Mask: 255.255.255.0
   - Default Gateway: 192.168.1.253
5. Configure the management IP address of Switch S1 on **VLAN 1**:
   S1(config)# interface vlan 1
   S1(config-if)# ip address 192.168.1.253 255.255.255.0
   S1(config-if)# no shutdown

**In the actual Packet Tracer**

For PC0:

PC0 → Desktop → IP Configuration

Enter:

IP Address: 192.168.1.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.253

Do the same for PC1 with 192.168.1.20.

For the switch:

**Switch → CLI**

enable

configure terminal

interface vlan 1

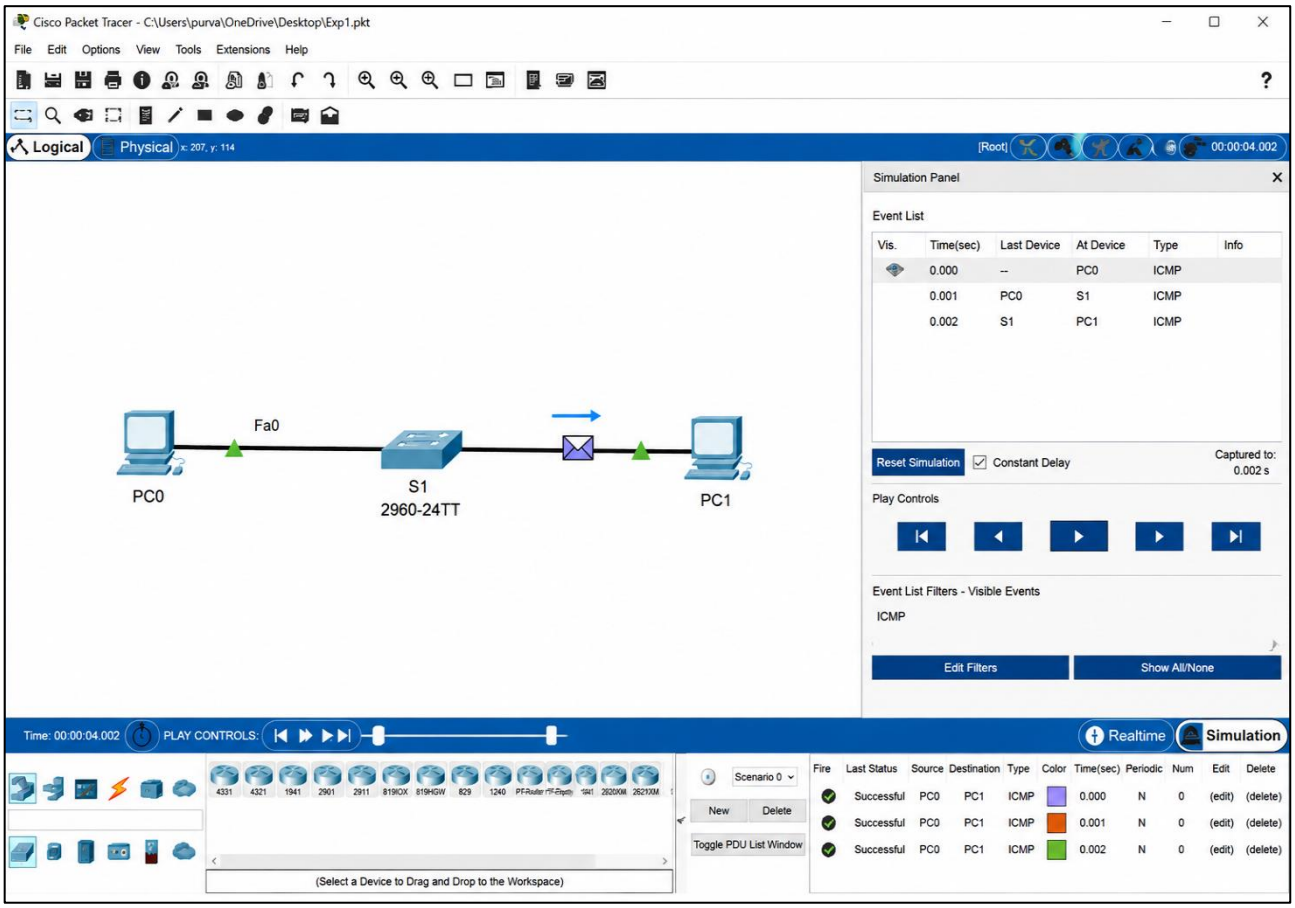
ip address 192.168.1.253 255.255.255.0

no shutdown

**Output**

The switch was successfully configured with basic security settings. The following configurations were verified:

- Hostname S1 configured successfully.
- Enable password and enable secret configured.
- Password encryption enabled.
- MOTD banner configured.
- Console password configured.
- VTY password configured.
- Switch management interface configured on VLAN 1.
- PC-to-switch connectivity verified successfully using ping.
- The running configuration was verified using show running-config.




**2. Switch Configuration and Security Implementation using Cisco Packet**

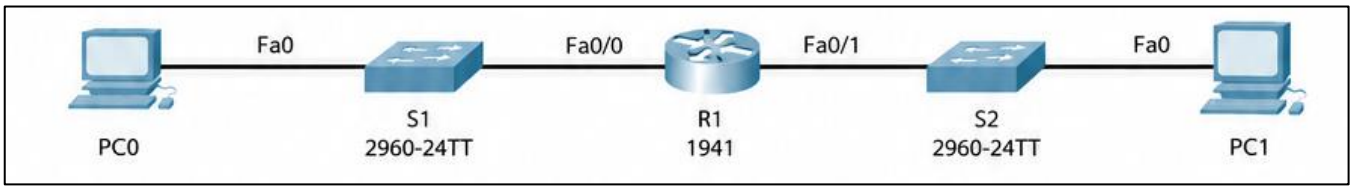
**A small organization is setting up its network to enable communication between in the different departments and external networks. A router is required to interconnect networks and manage data routing. The network administrator uses Cisco Packet Tracer to configure the router with basic commands and verify connectivity.**

**Aim**: - To configure a Cisco switch with basic security settings using Cisco Packet Tracer and verify the configuration.

**Requirements**

- Cisco Router
- Cisco Switches
- PCs
- Copper Straight-Through Cables

**Network Topology**




| Device | Interface | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|---|
| PC0 | FastEthernet0 | 192.168.1.10 | 255.255.255.0 | 192.168.1.1 |
| PC1 | FastEthernet0 | 192.168.2.10 | 255.255.255.0 | 192.168.2.1 |
| Router R1 | GigabitEthernet0/0 | 192.168.1.1 | 255.255.255.0 | — |
| Router R1 | GigabitEthernet0/1 | 192.168.2.1 | 255.255.255.0 | — |

**Procedure**

1. Open Cisco Packet Tracer and place one Cisco router, two switches, and two PCs in the logical workspace.
2. Connect PC0 to Switch S1 and PC1 to Switch S2 using Copper Straight-Through cables.
3. Connect Switch S1 to the router interface GigabitEthernet0/0.
4. Connect Switch S2 to the router interface GigabitEthernet0/1.
5. Configure the IP address of PC0:
   - IP Address: 192.168.1.10
   - Subnet Mask: 255.255.255.0
   - Default Gateway: 192.168.1.1
6. Configure the IP address of PC1:
   - IP Address: 192.168.2.10
   - Subnet Mask: 255.255.255.0
   - Default Gateway: 192.168.2.1
7. Configure the router. Select:

Router → CLI

8. Enter privileged EXEC mode:

Router> enable

9. Enter global configuration mode:

Router# configure terminal

10. Assign a hostname to the router:

```
Router(config)# hostname R1
```

11. Configure the first router interface:

```
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
```

12. Configure the second router interface:

```
R1(config)# interface gigabitEthernet 0/1
R1(config-if)# ip address 192.168.2.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
```

13. Configure a privileged EXEC password:

```
R1(config)# enable secret cisco
```

14. Configure a login warning message:

```
R1(config)# banner motd #Authorized Access Only#
```

15. Configure the console password:

```
R1(config)# line console 0
R1(config-line)# password cisco
R1(config-line)# login
R1(config-line)# exit
```

16. Configure VTY access:

```
R1(config)# line vty 0 4
R1(config-line)# password cisco
R1(config-line)# login
R1(config-line)# exit
```

17. Save the configuration:

```
R1(config)# end
R1# copy running-config startup-config
```

18. Verify the configured interfaces using:

```
R1# show ip interface brief
```

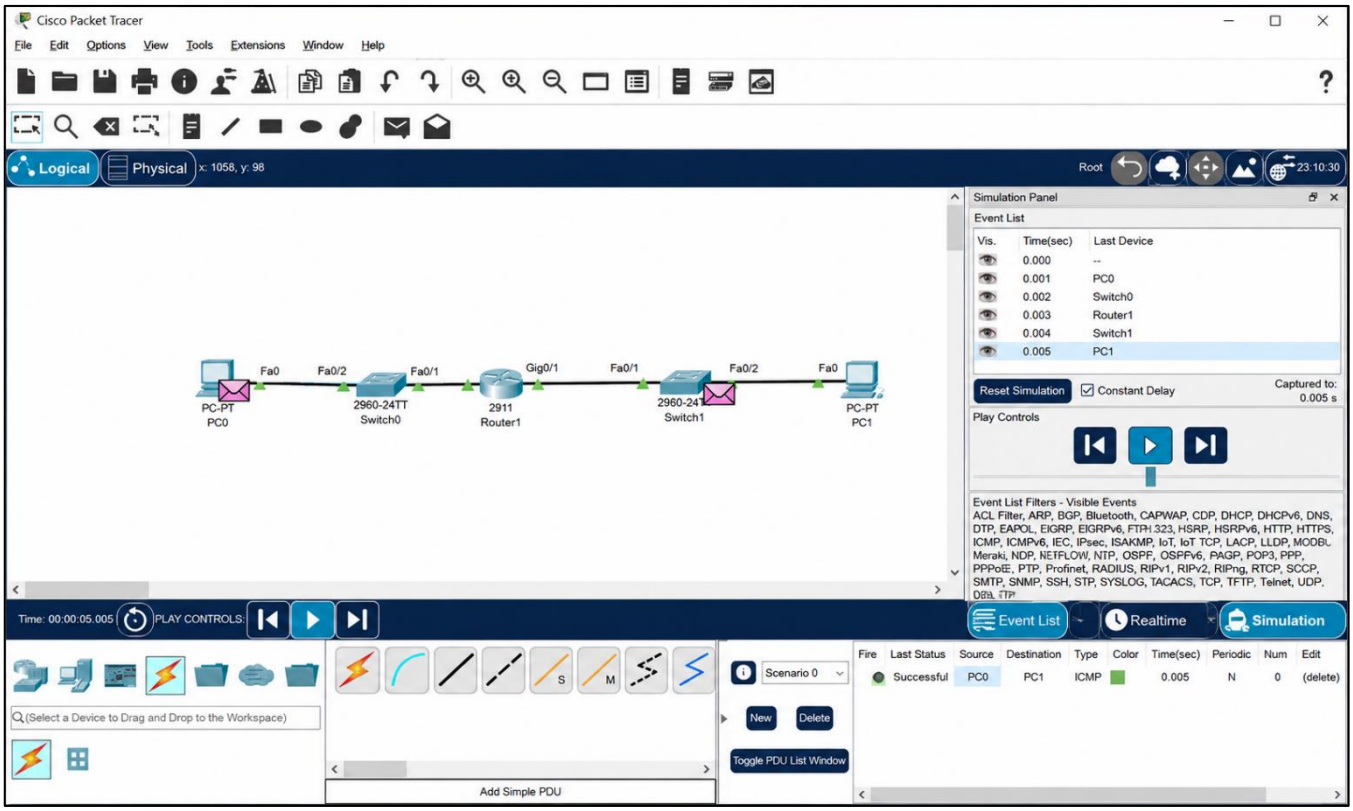
19. Verify the running configuration using:

R1# show running-config

20. From PC0, open Desktop → Command Prompt and test connectivity to PC1 using:

ping 192.168.2.10

**Output**



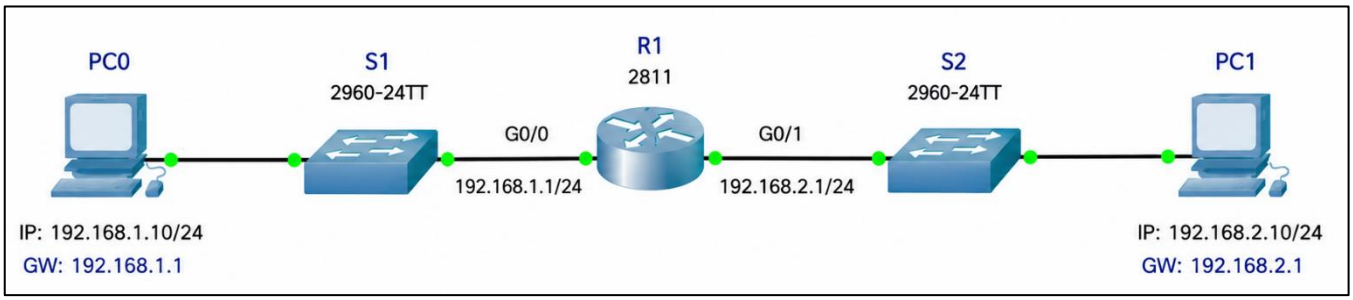

3. **IP Address Configuration and Default Routing using Cisco Packet Tracer.**
   **A company wants to connect its internal network to an external network (Internet/ISP). For communication beyond the local network, routers must be configured with IP addresses and a default route. The network administrator uses Cisco Packet Tracer to simulate and configure this setup.**

**Aim**: - To configure IP addresses on router interfaces and set a default route to enable communication between the internal network and external network using Cisco Packet Tracer.

**Requirements**

- Cisco Router
- Cisco Switches
- PCs
- Copper Straight-Through Cables

**Network Topology**




## IP ADDRESS CONFIGURATION TABLE

| Device | Interface | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|---|
| PC0 | FastEthernet0 | 192.168.1.10 | 255.255.255.0 | 192.168.1.1 |
| PC1 | FastEthernet0 | 192.168.2.10 | 255.255.255.0 | 192.168.2.1 |
| Router R1 | GigabitEthernet0/0 | 192.168.1.1 | 255.255.255.0 | - |
| Router R1 | GigabitEthernet0/1 | 192.168.2.1 | 255.255.255.0 | - |

**Procedure**

1. Open Cisco Packet Tracer and place one Cisco router, two switches, and two PCs in the workspace.
2. Connect PC0 to Switch S1 using Copper Straight-Through cable.
3. Connect Switch S1 to Router R1 interface GigabitEthernet0/0.
4. Connect Router R1 interface GigabitEthernet0/1 to Switch S2.
5. Connect Switch S2 to PC1 using Copper Straight-Through cable.
6. Configure the IP address of PC0:
    - IP Address: 192.168.1.10
    - Subnet Mask: 255.255.255.0
    - Default Gateway: 192.168.1.1
7. Configure the IP address of PC1:
    - IP Address: 192.168.2.10
    - Subnet Mask: 255.255.255.0
    - Default Gateway: 192.168.2.1

8. Configure the router. Select:

Router → CLI

9. Enter privileged EXEC mode:

Router> enable

10. Enter global configuration mode:

Router# configure terminal

11. Assign a hostname to the router:

Router(config)# hostname R1

12. Configure the first router interface:

R1(config)# interface gigabitEthernet 0/0

R1(config-if)# ip address 192.168.1.1 255.255.255.0

R1(config-if)# no shutdown

R1(config-if)# exit

13. Configure the second router interface:

R1(config)# interface gigabitEthernet 0/1

R1(config-if)# ip address 192.168.2.1 255.255.255.0

R1(config-if)# no shutdown

R1(config-if)# exit

14. Configure a privileged EXEC password:

R1(config)# enable secret cisco

15. Configure a login warning message:

R1(config)# banner motd #Authorized Access Only#

16. Configure the console password:

R1(config)# line console 0

R1(config-line)# password cisco

R1(config-line)# login

R1(config-line)# exit

17. Configure VTY access:

R1(config)# line vty 0 4

R1(config-line)# password cisco

R1(config-line)# login

R1(config-line)# exit

18. Save the configuration:

R1(config)# end

R1# copy running-config startup-config

19. Verify the configured interfaces using:

R1# show ip interface brief

20. Verify the running configuration using:

R1# show running-config

21. From PC0, open Desktop → Command Prompt and test connectivity to PC1 using:

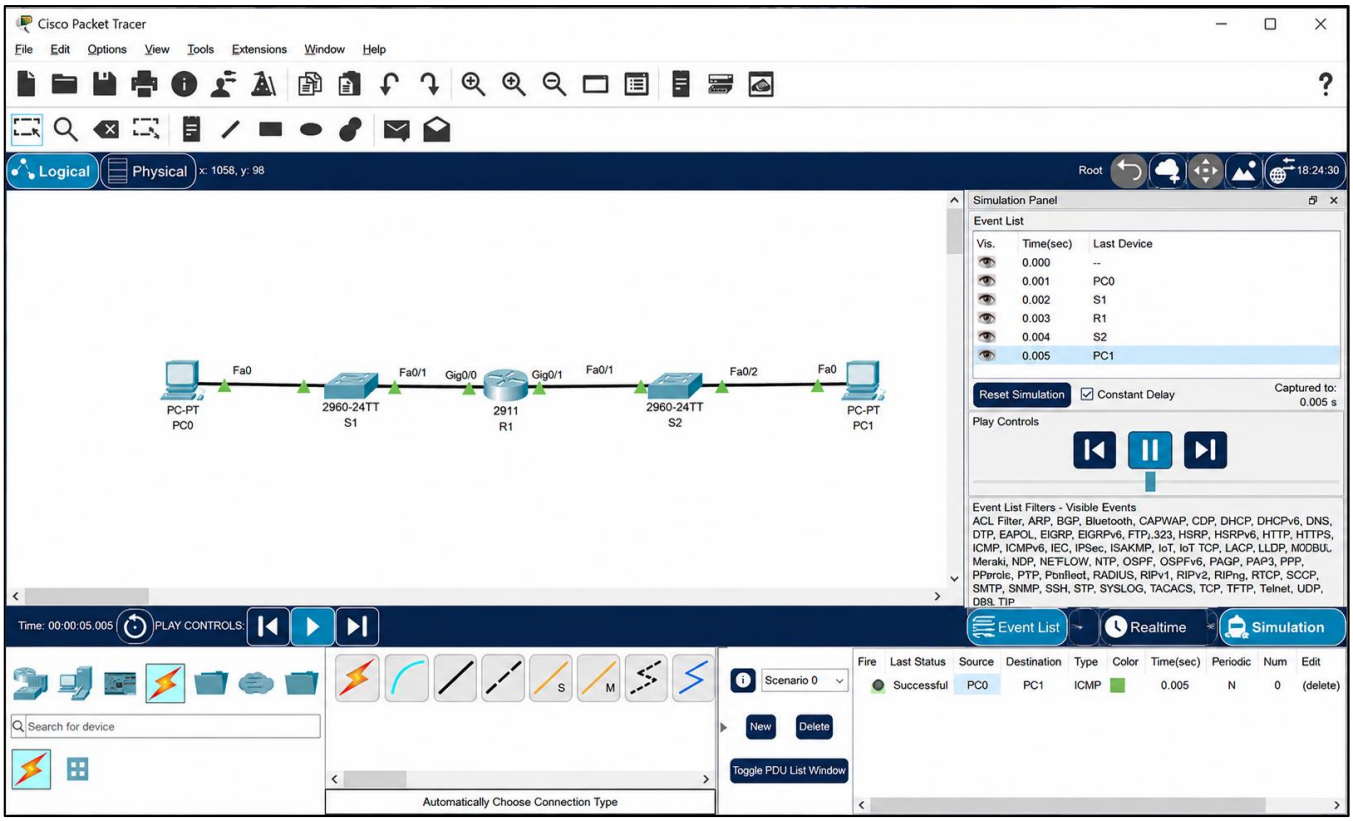
ping 192.168.2.10

22. If the ping is successful, communication between the two different networks has been established successfully.

**Output**

The router was successfully configured with IP addresses on both Giga bit Ethernet interfaces. The ICMP packet was successfully transmitted from **PC0 to PC1 through R1**, as verified in Simulation Mode.

The successful PDU event confirms communication between the two different networks.



**4. Configuration of Static Routing using Cisco Packet Tracer.**

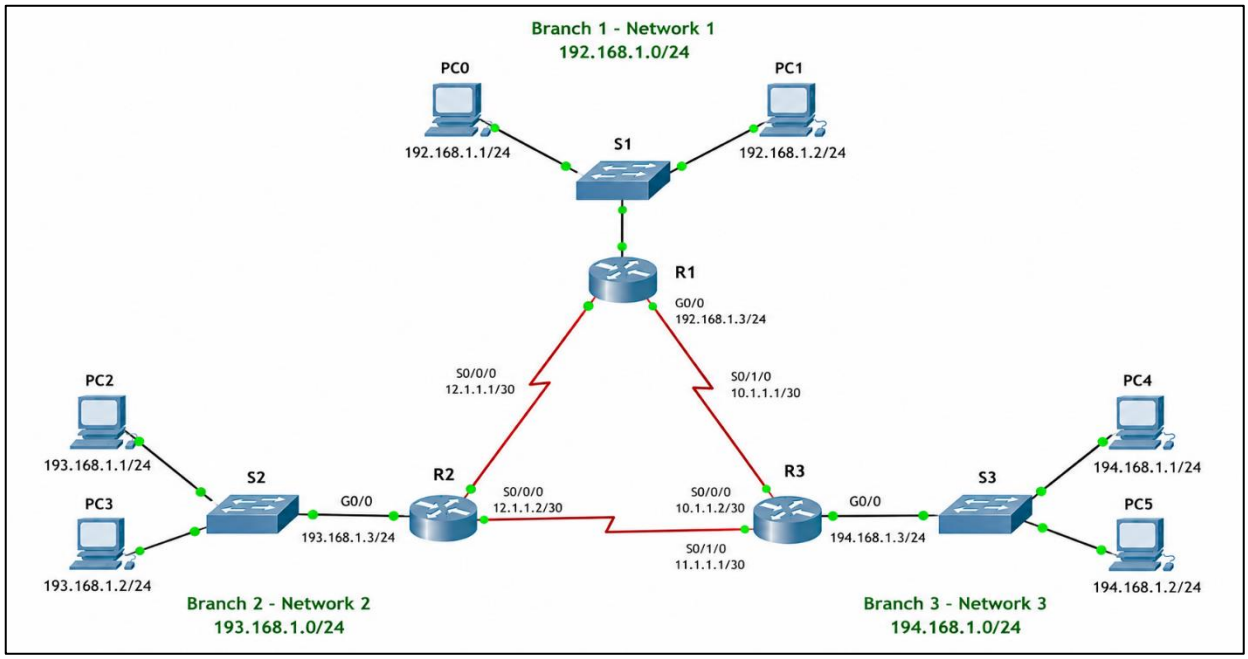
**A company has multiple branch networks that need to communicate with each other. Since the network is small and stable, the administrator decides to use static routing to manually define paths between networks. The configuration and testing are performed using Cisco Packet Tracer.**

**Aim**: - To configure static routing between multiple networks using Cisco Packet Tracer and verify communication between different networks.

**Apparatus Required**

- Cisco Packet Tracer
- Routers (3)
- Switches (3)
- PCs (6)
- Copper Straight-Through Cables
- Serial Cables

**Network Topology**




**Procedure**

1. Open Cisco Packet Tracer and place three routers, three switches, and six PCs in the logical workspace.

2. Connect the PCs to the switches using Copper Straight-Through cables.
3. Connect each switch to its corresponding router.
4. Connect the three routers using serial connections to form the required topology.
5. Select PC → Desktop → IP Configuration and configure the PCs with the IP addresses given in the reference manual:
    - PC0 → 192.168.1.1
    - PC1 → 192.168.1.2
    - PC2 → 193.168.1.1
    - PC3 → 193.168.1.2
    - PC4 → 194.168.1.1
    - PC5 → 194.168.1.2
6. Configure the default gateways:
    - PC0 and PC1 → 192.168.1.3
    - PC2 and PC3 → 193.168.1.3
    - PC4 and PC5 → 194.168.1.3
7. Configure the router-to-router connections.

Router0 to Router1:

Router0 → Config → Serial3/0 → 12.1.1.1

Router1 → Config → Serial3/0 → 12.1.1.2

Router0 to Router2:

Router0 → Config → Serial2/0 → 10.1.1.1

Router2 → Config → Serial2/0 → 10.1.1.2

Router1 to Router2:

Router1 → Config → Serial2/0 → 11.1.1.2

Router2 → Config → Serial3/0 → 11.1.1.1

These interface assignments follow the uploaded reference manual.

8. Configure static routing on Router0.

Route towards the network connected to Router1:

Router0 → Config → Static

Network: 194.168.1.0

Subnet Mask: 255.255.255.0

Next Hop: 12.1.1.2

Route towards the network connected to Router2:

Router0 → Config → Static

Network: 193.168.1.0

Subnet Mask: 255.255.255.0

Next Hop: 10.1.1.2

9. Configure static routing on Router1.

Route towards Router0:

Router1 → Config → Static

Network: 192.168.1.0

Subnet Mask: 255.255.255.0

Next Hop: 12.1.1.1

Route towards Router2:

Router1 → Config → Static

Network: 193.168.1.0

Subnet Mask: 255.255.255.0

Next Hop: 11.1.1.1

10. Configure static routing on Router2.

Router2 → Config → Static

Network: 194.168.1.0

Subnet Mask: 255.255.255.0

Next Hop: 11.1.1.2

Router2 → Config → Static

Network: 194.168.1.0

Subnet Mask: 255.255.255.0

Next Hop: 10.1.1.1

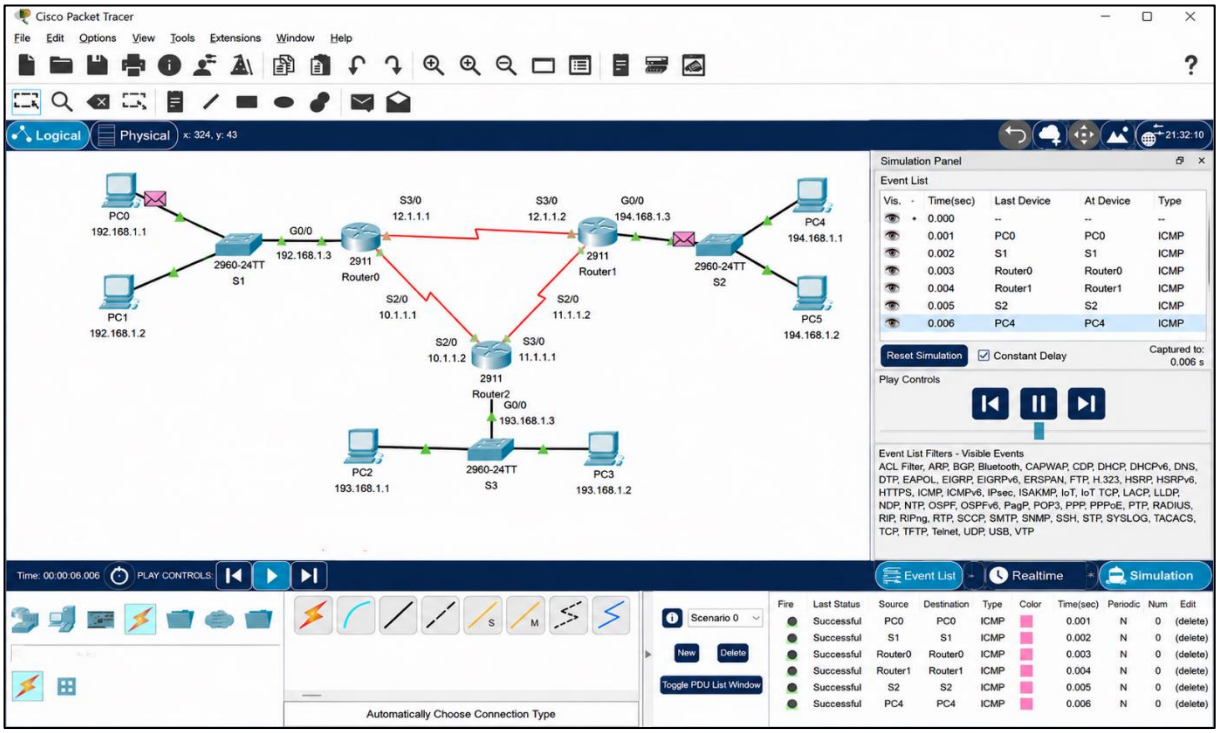
These static-route entries reproduce the configuration shown in the reference manual.

11. Configure the IP addresses obtained from the network topology on all the PCs and router interfaces.
12. Save the router configurations.
13. Pass a simple PDU from one network to another to check whether the packet is delivered successfully.
14. Open the Simulation environment in Cisco Packet Tracer to observe the path followed by the packet as it moves from one network to another.

## IP ADDRESS TABLE

| Device | Interface | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|---|
| PC0 | FastEthernet0 | 192.168.1.1 | 255.255.255.0 | 192.168.1.3 |
| PC1 | FastEthernet0 | 192.168.1.2 | 255.255.255.0 | 192.168.1.3 |
| PC2 | FastEthernet0 | 193.168.1.1 | 255.255.255.0 | 193.168.1.3 |
| PC3 | FastEthernet0 | 193.168.1.2 | 255.255.255.0 | 193.168.1.3 |
| PC4 | FastEthernet0 | 194.168.1.1 | 255.255.255.0 | 194.168.1.3 |
| PC5 | FastEthernet0 | 194.168.1.2 | 255.255.255.0 | 194.168.1.3 |
| Router R1 | GigabitEthernet0/0 | 192.168.1.3 | 255.255.255.0 | - |
| Router R1 | Serial0/0/0 | 12.1.1.1 | 255.255.255.252 | - |
| Router R1 | Serial0/1/0 | 10.1.1.1 | 255.255.255.252 | - |
| Router R2 | GigabitEthernet0/0 | 193.168.1.3 | 255.255.255.0 | - |
| Router R2 | Serial0/0/0 | 12.1.1.2 | 255.255.255.252 | - |
| Router R2 | Serial0/1/0 | 11.1.1.2 | 255.255.255.252 | - |
| Router R3 | GigabitEthernet0/0 | 194.168.1.3 | 255.255.255.0 | - |
| Router R3 | Serial0/0/0 | 10.1.1.2 | 255.255.255.252 | - |
| Router R3 | Serial0/1/0 | 11.1.1.1 | 255.255.255.252 | - |

**Output**



## 5. Configuration of Dynamic Routing using Cisco Packet Tracer

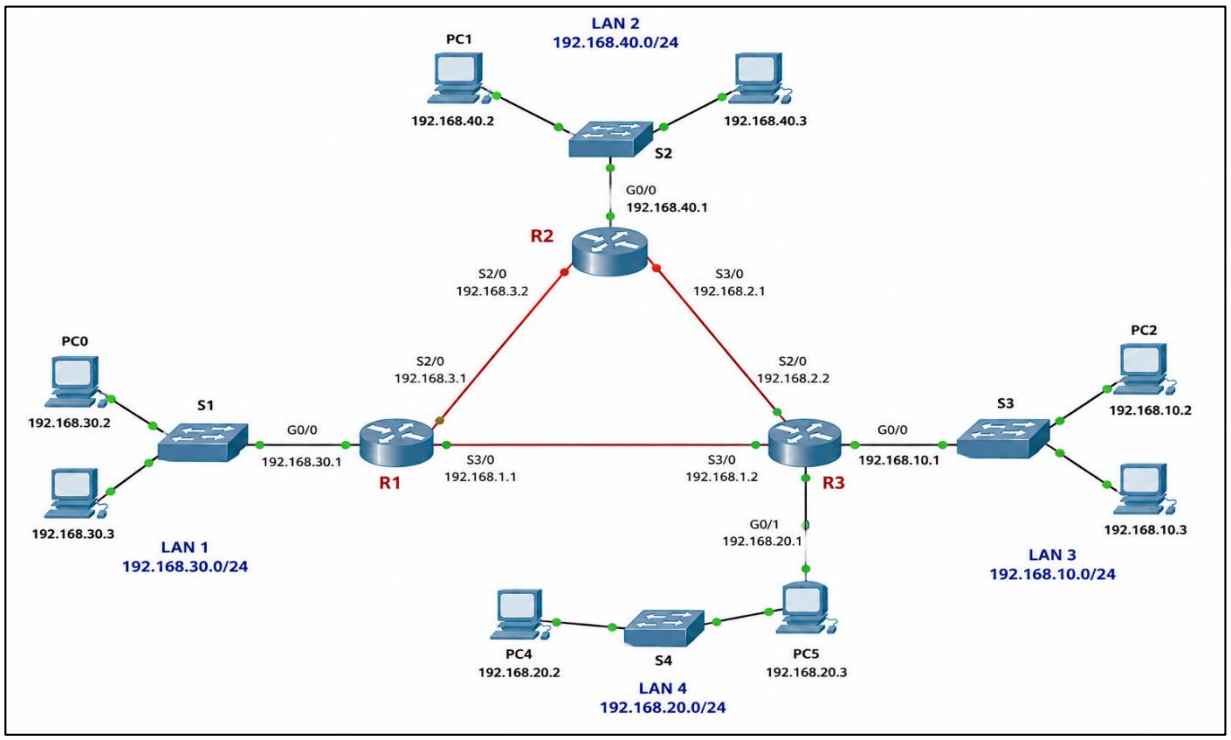
**A growing organization has multiple interconnected networks. Manually configuring routes is complex and error-prone, so the network administrator adopts dynamic routing, where routers automatically exchange routing information. The implementation and testing are performed using Cisco Packet Tracer.**

**Aim**: - To configure dynamic routing between multiple interconnected networks using Cisco Packet Tracer and verify communication between different networks.

### Apparatus Required

- Cisco Packet Tracer
- Routers (3)
- Switches (3)
- PCs (6)
- Copper Straight-Through Cables
- Serial Cables

### Network Topology



- PC0 → 192.168.30.2
- PC1 → 192.168.40.2
- PC2 → 192.168.10.2
- PC3 → 192.168.10.3
- PC4 → 192.168.20.2
- PC5 → 192.168.20.3

Router-to-router links:

- R1–R2 → 192.168.3.1 / 192.168.3.2
- R1–R3 → 192.168.1.1 / 192.168.1.2
- R2–R3 → 192.168.2.1 / 192.168.2.2

**IP ADDRESS TABLE**

| Device | Interface | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|---|
| PC0 | FastEthernet0 | 192.168.30.2 | 255.255.255.0 | 192.168.30.1 |
| PC1 | FastEthernet0 | 192.168.40.2 | 255.255.255.0 | 192.168.40.1 |
| PC2 | FastEthernet0 | 192.168.10.2 | 255.255.255.0 | 192.168.10.1 |
| PC3 | FastEthernet0 | 192.168.10.3 | 255.255.255.0 | 192.168.10.1 |
| PC4 | FastEthernet0 | 192.168.20.2 | 255.255.255.0 | 192.168.20.1 |
| PC5 | FastEthernet0 | 192.168.20.3 | 255.255.255.0 | 192.168.20.1 |
| R1 | GigabitEthernet0/0 | 192.168.30.1 | 255.255.255.0 | - |
| R1 | Serial2/0 | 192.168.3.1 | 255.255.255.0 | - |
| R1 | Serial3/0 | 192.168.1.1 | 255.255.255.0 | - |
| R2 | GigabitEthernet0/0 | 192.168.40.1 | 255.255.255.0 | - |
| R2 | Serial2/0 | 192.168.3.2 | 255.255.255.0 | - |
| R2 | Serial3/0 | 192.168.2.1 | 255.255.255.0 | - |
| R3 | GigabitEthernet0/0 | 192.168.10.1 | 255.255.255.0 | - |
| R3 | GigabitEthernet0/1 | 192.168.20.1 | 255.255.255.0 | - |
| R3 | Serial2/0 | 192.168.2.2 | 255.255.255.0 | - |
| R3 | Serial3/0 | 192.168.1.2 | 255.255.255.0 | - |

**Procedure**

1. Open Cisco Packet Tracer and place three routers, three switches, and six PCs in the logical workspace.
2. Connect the PCs to the respective switches using Copper Straight-Through cables.

3. Connect each switch to its corresponding router.
4. Connect the three routers using serial connections according to the network topology.
5. Configure the IP address of the PCs:
    - PC0 → 192.168.30.2
    - PC1 → 192.168.40.2
    - PC2 → 192.168.10.2
    - PC3 → 192.168.10.3
    - PC4 → 192.168.20.2
    - PC5 → 192.168.20.3
6. Configure the default gateways:
    - PC0 → 192.168.30.1
    - PC1 → 192.168.40.1
    - PC2 and PC3 → 192.168.10.1
    - PC4 and PC5 → 192.168.20.1
7. Configure the router-to-router connections.

    Router0 to Router1:

    Router0 → Config → Serial2/0 → 192.168.3.1

    Router1 → Config → Serial2/0 → 192.168.3.2

    Router0 to Router2:

    Router0 → Config → Serial3/0 → 192.168.1.1

    Router2 → Config → Serial3/0 → 192.168.1.2

    Router1 to Router2:

    Router1 → Config → Serial3/0 → 192.168.2.1

    Router2 → Config → Serial2/0 → 192.168.2.2

    These connections and interface addresses are given in the reference manual.
8. Configure RIP on Router0.

    Router0 → Config → RIP

    Add the directly connected networks:

    192.168.1.0

    192.168.3.0

    192.168.30.0
9. Configure RIP on Router1.

    Router1 → Config → RIP

Add the directly connected networks:

192.168.2.0

192.168.3.0

192.168.40.0

10. Configure RIP on Router2.

    Router2 → Config → RIP

    Add the directly connected networks:

    192.168.1.0

    192.168.2.0

    192.168.10.0

    192.168.20.0

    The reference manual specifies these directly connected networks for RIP configuration on the three routers.

11. Save the router configurations.
12. Allow the routers to exchange routing information using RIP.
13. Pass a simple PDU from one network to another and verify that the packet is successfully communicated.
14. Open the Simulation environment in Cisco Packet Tracer to observe the path followed by the packet between different networks.

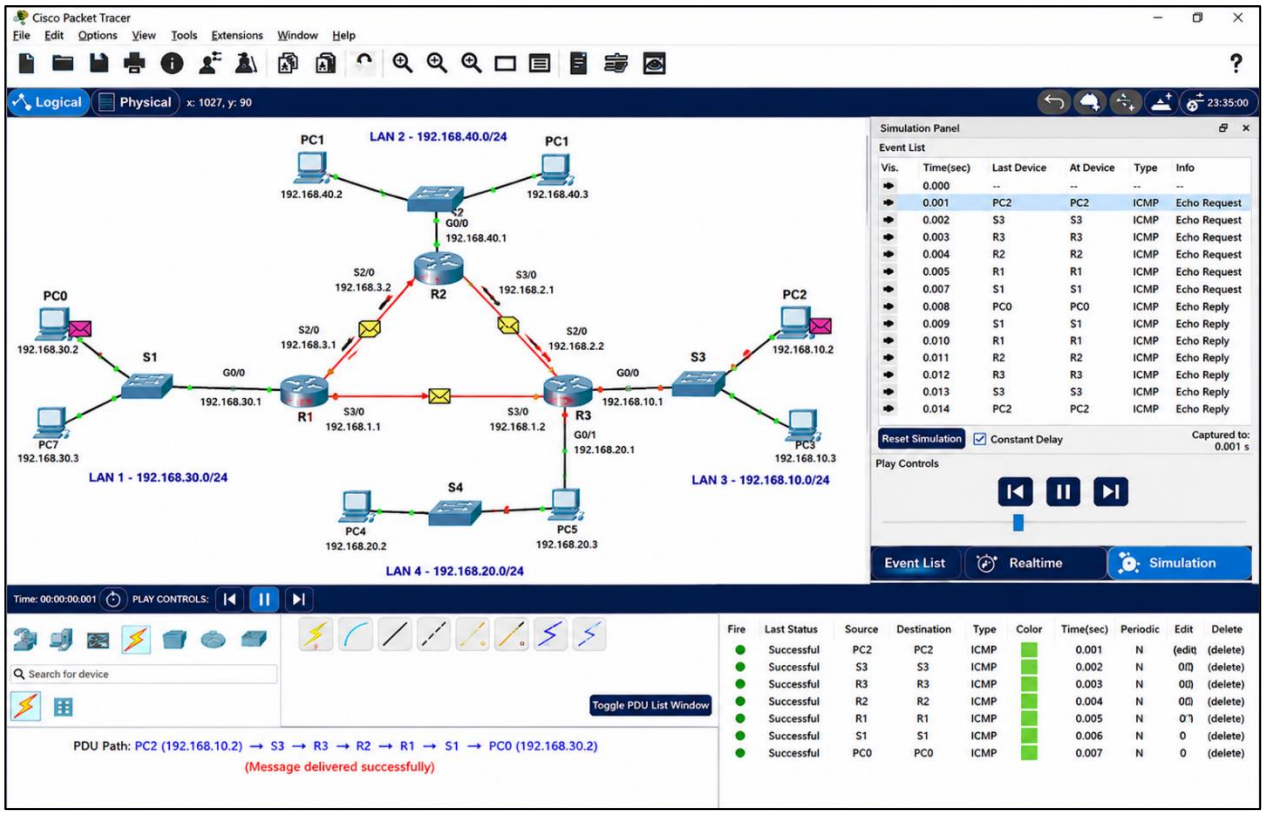
**Output**

The dynamic routing configuration was successfully implemented using RIP on all three routers. The routers exchanged routing information and dynamically learned the remote networks.

The ICMP packet was successfully transmitted from **PC2 (192.168.10.2)** to **PC0 (192.168.30.2)** through the path:

**PC2 → S3 → R3 → R2 → R1 → S1 → PC0**

The packet path and successful delivery were verified using **Simulation Mode** in Cisco Packet Tracer.




## 6. Implementation of EIGRP using Cisco Packet Tracer

As an organization expands, it requires a fast, scalable, and efficient routing protocol. To achieve quick convergence and optimal path selection, the network administrator implements EIGRP. The configuration and testing are performed using Cisco Packet Tracer.

**Aim**: - To configure EIGRP on multiple routers using Cisco Packet Tracer and verify dynamic routing, fast convergence, and connectivity between different networks.

### Apparatus Required

- Cisco Packet Tracer
- Routers (3)
- Switches (3)
- PCs (6)
- Copper Straight-Through Cables

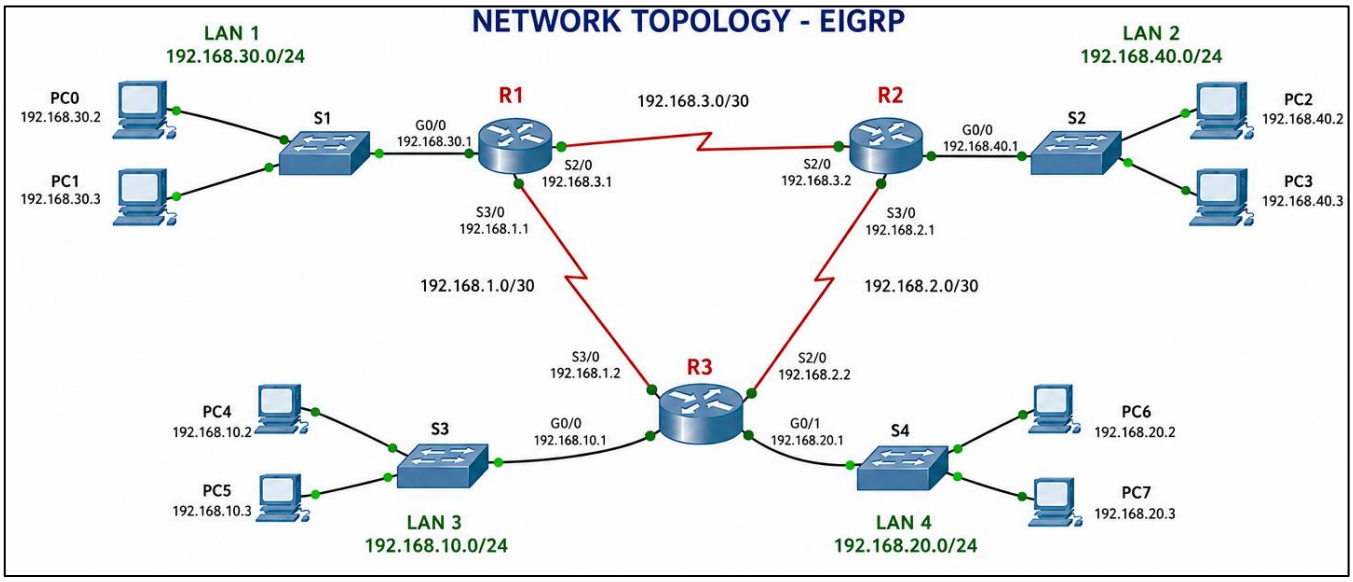
### Network Addressing

- Network 1: 192.168.30.0/24
- Network 2: 192.168.40.0/24

- Network 3: 192.168.10.0/24
- Network 4: 192.168.20.0/24

Router interconnections:

- R1–R2: 192.168.3.0/30
- R1–R3: 192.168.1.0/30
- R2–R3: 192.168.2.0/30




**Procedure**

1. Open Cisco Packet Tracer and place three routers, three switches, and six PCs in the logical workspace.
2. Connect the PCs to their respective switches using Copper Straight-Through cables.
3. Connect each switch to its corresponding router.
4. Connect the routers using serial cables according to the network topology.
5. Configure the IP addresses of the PCs:
   - PC0 → 192.168.30.2
   - PC1 → 192.168.40.2
   - PC2 → 192.168.10.2
   - PC3 → 192.168.10.3
   - PC4 → 192.168.20.2
   - PC5 → 192.168.20.3
6. Configure the default gateways:
   - PC0 → 192.168.30.1

- PC1 → 192.168.40.1
- PC2 and PC3 → 192.168.10.1
- PC4 and PC5 → 192.168.20.1

7. Configure the router interfaces according to the network topology.
8. On Router R1, enter the CLI and configure EIGRP:

```
R1> enable
R1# configure terminal
R1(config)# router eigrp 100
R1(config-router)# network 192.168.30.0
R1(config-router)# network 192.168.3.0
R1(config-router)# network 192.168.1.0
R1(config-router)# no auto-summary
R1(config-router)# exit
```

9. On Router R2, configure EIGRP:

```
R2> enable
R2# configure terminal
R2(config)# router eigrp 100
R2(config-router)# network 192.168.40.0
R2(config-router)# network 192.168.3.0
R2(config-router)# network 192.168.2.0
R2(config-router)# no auto-summary
R2(config-router)# exit
```

10. On Router R3, configure EIGRP:

```
R3> enable
R3# configure terminal
R3(config)# router eigrp 100
R3(config-router)# network 192.168.10.0
R3(config-router)# network 192.168.20.0
R3(config-router)# network 192.168.1.0
R3(config-router)# network 192.168.2.0
R3(config-router)# no auto-summary
R3(config-router)# exit
```

11. Save the configurations:

R1# copy running-config startup-config

R2# copy running-config startup-config

R3# copy running-config startup-config

12. Verify the EIGRP neighbor relationships:

    R1# show ip eigrp neighbors

    R2# show ip eigrp neighbors

    R3# show ip eigrp neighbors

13. Verify the routing table:

    R1# show ip route

    R2# show ip route

    R3# show ip route

14. Verify EIGRP routes in the routing table. EIGRP-learned routes are identified by the D code.
15. From PC0, open Desktop → Command Prompt and test connectivity to a PC in another network:

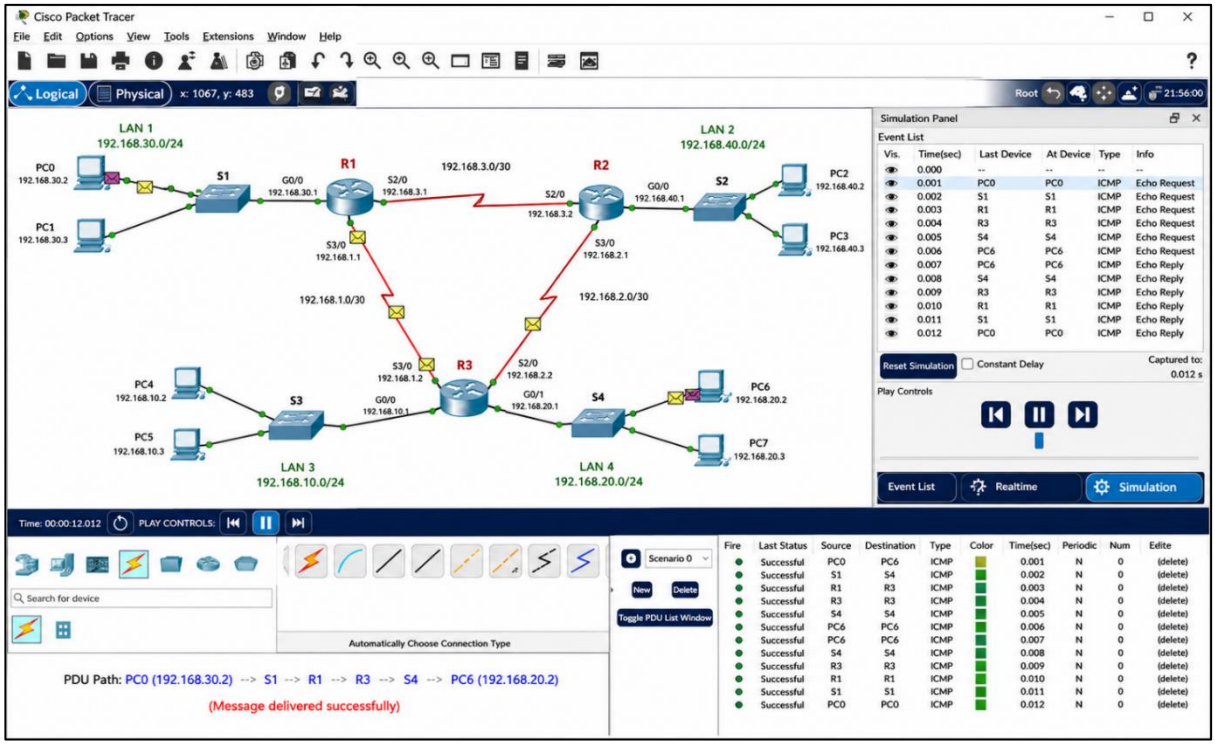
    PC> ping 192.168.20.2

16. Similarly, test communication between other networks to verify that routing information has been exchanged successfully.
17. Open the Simulation environment and observe the path followed by packets between the different networks.

**IP ADDRESS TABLE - EIGRP**

| Device | Interface | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|---|
| PC0 | FastEthernet0 | 192.168.30.2 | 255.255.255.0 | 192.168.30.1 |
| PC1 | FastEthernet0 | 192.168.30.3 | 255.255.255.0 | 192.168.30.1 |
| PC2 | FastEthernet0 | 192.168.40.2 | 255.255.255.0 | 192.168.40.1 |
| PC3 | FastEthernet0 | 192.168.40.3 | 255.255.255.0 | 192.168.40.1 |
| PC4 | FastEthernet0 | 192.168.10.2 | 255.255.255.0 | 192.168.10.1 |
| PC5 | FastEthernet0 | 192.168.10.3 | 255.255.255.0 | 192.168.10.1 |
| R1 | GigabitEthernet0/0 | 192.168.30.1 | 255.255.255.0 | - |
| R1 | Serial2/0 | 192.168.3.1 | 255.255.255.252 | - |
| R1 | Serial3/0 | 192.168.1.1 | 255.255.255.252 | - |
| R2 | GigabitEthernet0/0 | 192.168.40.1 | 255.255.255.0 | - |
| R2 | Serial2/0 | 192.168.3.2 | 255.255.255.252 | - |
| R2 | Serial3/0 | 192.168.2.1 | 255.255.255.252 | - |
| R3 | GigabitEthernet0/0 | 192.168.10.1 | 255.255.255.0 | - |
| R3 | GigabitEthernet0/1 | 192.168.20.1 | 255.255.255.0 | - |
| R3 | Serial2/0 | 192.168.2.2 | 255.255.255.252 | - |
| R3 | Serial3/0 | 192.168.1.2 | 255.255.255.252 | - |

## Output




7. **VLAN Configuration, Inter-VLAN Routing, VTP & Switch Troubleshooting using Cisco Packet Tracer.**

   **A company wants to segment its network into different departments (HR, Finance, IT) for better security and performance. To achieve this, the network administrator implements VLANs, enables communication between them using inter-VLAN routing, manages VLANs centrally using VTP, and performs troubleshooting. The setup is simulated using Cisco Packet Tracer.**

   **Aim**: - To configure VLANs for different departments, implement Inter-VLAN Routing, manage VLAN information using VTP, and troubleshoot switch configuration using Cisco Packet Tracer.

   **Apparatus Required**

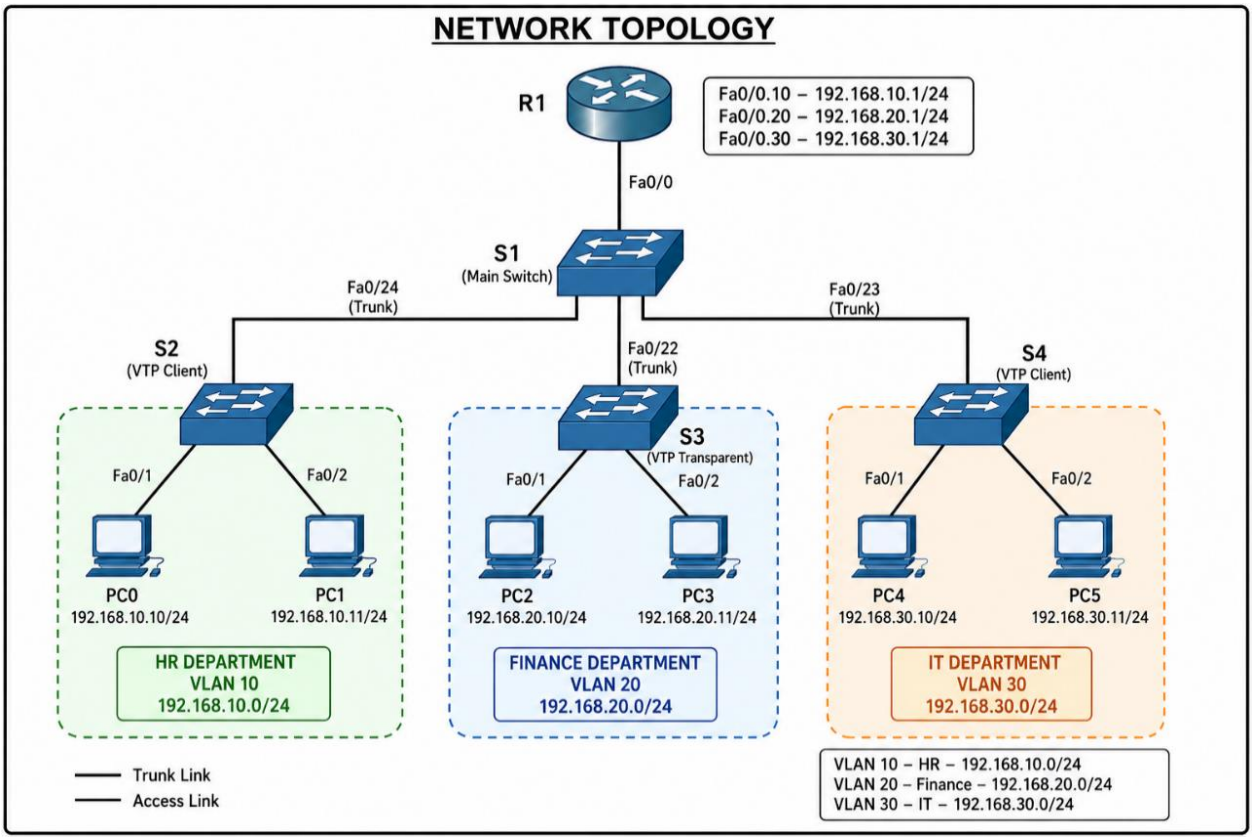
   - Cisco Packet Tracer
   - Routers (1)
   - Switches (3)
   - PCs (6)
   - Copper Straight-Through Cables

**Network Topology**

The network is divided into three departments:

- HR → VLAN 10 → 192.168.10.0/24
- Finance → VLAN 20 → 192.168.20.0/24
- IT → VLAN 30 → 192.168.30.0/24

The switches are connected through trunk links, and Router R1 provides communication between the different VLANs.



## IP ADDRESS CONFIGURATION

| Device | Interface | VLAN | IP Address | Subnet Mask | Default Gateway | Description |
|---|---|---|---|---|---|---|
| HR (VLAN 10) 192.168.10.0/24 | PC0 | NIC | 192.168.10.10 | 255.255.255.0 | 192.168.10.1 | HR PC-1 |
| | PC1 | NIC | 192.168.10.11 | 255.255.255.0 | 192.168.10.1 | HR PC-2 |
| | R1 | Fa0/0.10 | 192.168.10.1 | 255.255.255.0 | - | Gateway for VLAN 10 |
| Finance (VLAN 20) 192.168.20.0/24 | PC2 | NIC | 192.168.20.10 | 255.255.255.0 | 192.168.20.1 | Finance PC-1 |
| | PC3 | NIC | 192.168.20.11 | 255.255.255.0 | 192.168.20.1 | Finance PC-2 |
| | R1 | Fa0/0.20 | 192.168.20.1 | 255.255.255.0 | - | Gateway for VLAN 20 |
| IT (VLAN 30) 192.168.30.0/24 | PC4 | NIC | 192.168.30.10 | 255.255.255.0 | 192.168.30.1 | IT PC-1 |
| | PC5 | NIC | 192.168.30.11 | 255.255.255.0 | 192.168.30.1 | IT PC-2 |
| | R1 | Fa0/0.30 | 192.168.30.1 | 255.255.255.0 | - | Gateway for VLAN 30 |
| Infrastructure Devices | S1 (Main Switch) | VLAN 1 | 192.168.1.1 | 255.255.255.0 | - | Management IP |
| | S2 (VTP Client) | VLAN 1 | 192.168.1.2 | 255.255.255.0 | 192.168.1.1 | Management IP |
| | S3 (VTP Transparent) | VLAN 1 | 192.168.1.3 | 255.255.255.0 | 192.168.1.1 | Management IP |
| | R1 (Router) | Fa0/0 | 192.168.1.254 | 255.255.255.0 | - | Management IP |
| Trunk Links (Inter-Switch & Router Link) | S1 Fa0/24 ↔ S2 Fa0/24 | Trunk Link (Allow VLANs 10, 20, 30) | | | | |
| | S1 Fa0/23 ↔ S3 Fa0/23 | Trunk Link (Allow VLANs 10, 20, 30) | | | | |
| | S1 Fa0/22 ↔ R1 Fa0/0 | Trunk Link (Allow VLANs 10, 20, 30) | | | | |

Note: All PCs use their respective VLAN gateway IP as Default Gateway.

**Procedure**

1. Open Cisco Packet Tracer and place one router, three switches, and six PCs in the logical workspace.
2. Connect the PCs to the switches using Copper Straight-Through cables.
3. Connect the switches to one another using trunk links.
4. Connect the main switch to Router R1.
5. Configure the PCs with the following IP addresses:

   HR – VLAN 10

   - PC0 → 192.168.10.10
   - PC1 → 192.168.10.11
   - Default Gateway → 192.168.10.1

   Finance – VLAN 20

   - PC2 → 192.168.20.10
   - PC3 → 192.168.20.11
   - Default Gateway → 192.168.20.1

   IT – VLAN 30

   - PC4 → 192.168.30.10

- PC5 → 192.168.30.11
- Default Gateway → 192.168.30.1

6. Configure VLAN 10, VLAN 20, and VLAN 30 on the main switch.

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name HR
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name Finance
Switch(config-vlan)# exit
Switch(config)# vlan 30
Switch(config-vlan)# name IT
Switch(config-vlan)# exit
```

The original manual demonstrates the same VLAN creation procedure using vlan 10, name VLAN-10, vlan 20, and name VLAN-20.

7. Assign the required access ports to the VLANs.

```
Switch(config)# interface range fa0/1-2
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit
Switch(config)# interface range fa0/3-4
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# exit
Switch(config)# interface range fa0/5-6
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 30
Switch(config-if-range)# exit
```

8. Configure the inter-switch links as trunk ports.

```
Switch(config)# interface fa0/24
Switch(config-if)# switchport mode trunk
Switch(config-if)# exit
```

The reference manual uses trunk mode on the switch interface for carrying VLAN traffic.

9. Configure Router R1 for Inter-VLAN Routing.

```
Router> enable
Router# configure terminal
Router(config)# interface fa0/0
Router(config-if)# no shutdown
Router(config-if)# exit
```

10. Configure the router sub-interface for VLAN 10.

```
Router(config)# interface fa0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# exit
```

11. Configure the router sub-interface for VLAN 20.

```
Router(config)# interface fa0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# exit
```

12. Configure the router sub-interface for VLAN 30.

```
Router(config)# interface fa0/0.30
Router(config-subif)# encapsulation dot1Q 30
Router(config-subif)# ip address 192.168.30.1 255.255.255.0
Router(config-subif)# exit
```

The uploaded manual demonstrates the same Router-on-a-Stick approach using 802.1Q encapsulation and VLAN-specific sub interfaces.

13. Configure VTP on the switches.

Switch S1 – VTP Server

```
S1> enable
S1# configure terminal
S1(config)# vtp mode server
S1(config)# exit
S1# show vtp status
```

Switch S2 – VTP Client

```
S2> enable
```

S2# configure terminal

S2(config)# vtp mode client

S2(config)# exit

S2# show vtp status

Switch S3 – VTP Transparent

S3> enable

S3# configure terminal

S3(config)# vtp mode transparent

S3(config)# exit

S3# show vtp status

The reference manual uses VTP Server, Client, and Transparent modes and verifies them using show vtp status.

14. Verify the VLAN configuration on the switches.

    Switch# show vlan brief

15. Verify the trunk configuration.

    Switch# show interfaces trunk

16. Verify the VTP configuration.

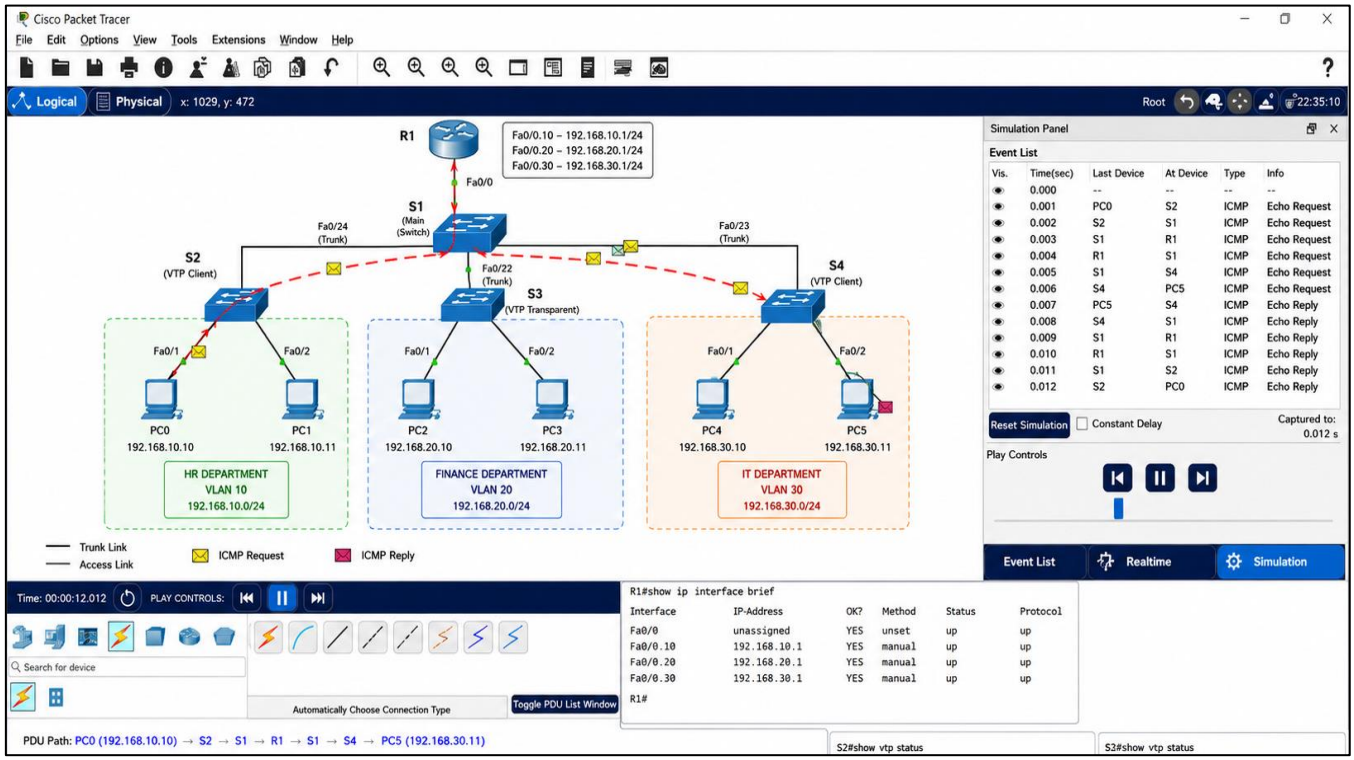
    Switch# show vtp status

17. Verify the router interfaces and sub interfaces.

    Router# show ip interface brief

18. Test connectivity between PCs belonging to the same VLAN using the ping command.
19. Test connectivity between PCs belonging to different VLANs using the ping command.
20. If communication fails, troubleshoot the configuration by checking:
    - Correct VLAN assignment of access ports.
    - Trunk configuration between switches.
    - VTP mode and VTP status.
    - Router sub interface configuration.
    - VLAN IDs used in encapsulation dot1Q.
    - IP addresses and default gateways.
    - Interface status using show ip interface brief.
21. After correcting any configuration errors, repeat the connectivity test.

**Output**




8. **Configuration of Standard Access Control Lists (ACLs) using Cisco Packet Tracer.**

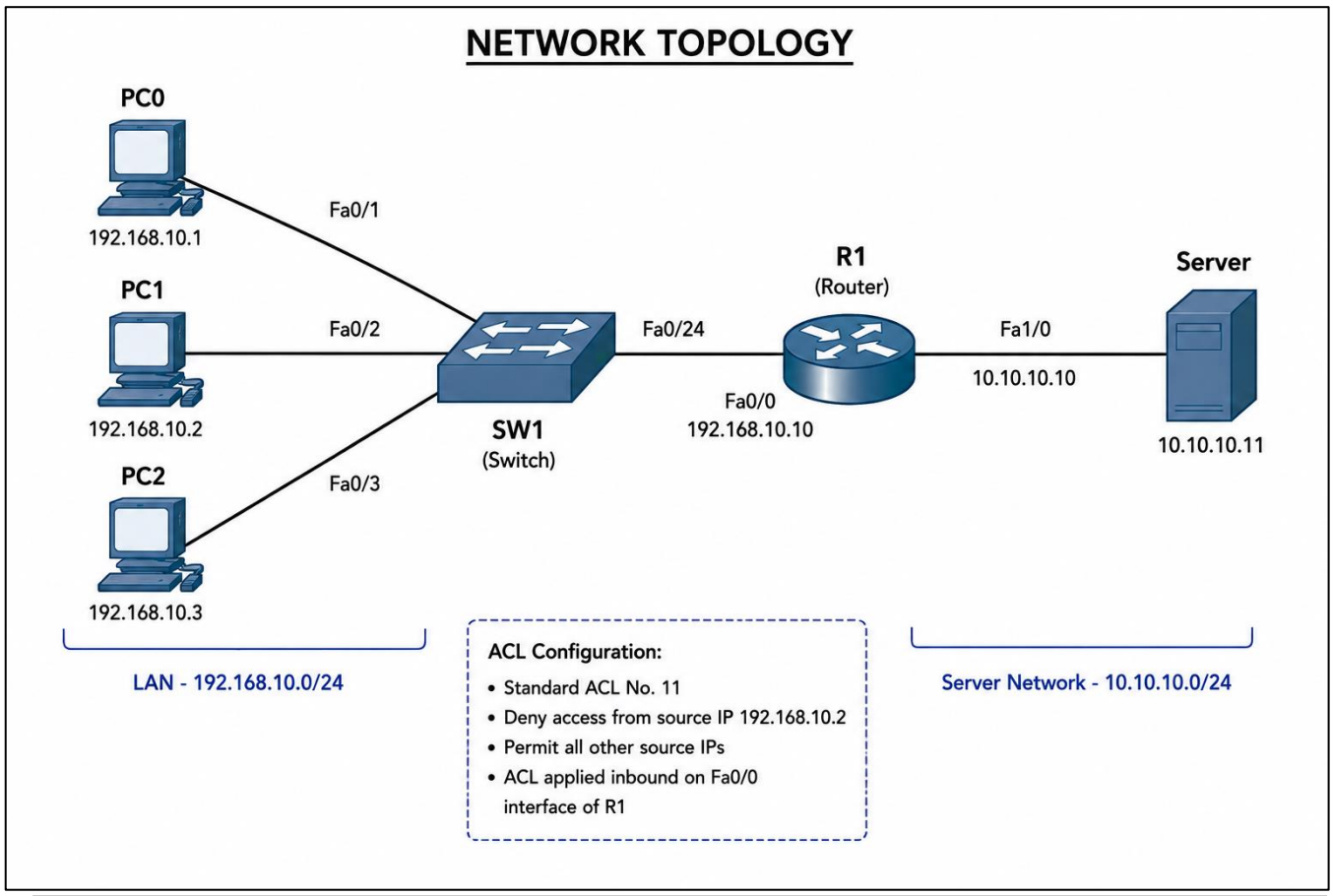
   **A company wants to restrict network access based only on source IP addresses. To implement basic security policies, the network administrator uses Standard Access Control Lists (ACLs) on routers. The configuration and testing are carried out using Cisco Packet Tracer.**

**Aim**: - To configure a Standard Access Control List (ACL) on a Cisco router to restrict network access based on source IP address and verify the access control policy using Cisco Packet Tracer.

**Apparatus Required**

1. Cisco Packet Tracer
2. Cisco Router
3. Cisco Switch
4. PCs (3)
5. Server
6. Copper Straight-Through Cables

**Network Topology**




## IP ADDRESS CONFIGURATION

| Device | Interface | IP Address | Subnet Mask | Default Gateway | Description |
|---|---|---|---|---|---|
| PC0 | FastEthernet0 | 192.168.10.1 | 255.255.255.0 | 192.168.10.10 | Client PC 0 |
| PC1 | FastEthernet0 | 192.168.10.2 | 255.255.255.0 | 192.168.10.10 | Client PC 1 (Blocked by ACL) |
| PC2 | FastEthernet0 | 192.168.10.3 | 255.255.255.0 | 192.168.10.10 | Client PC 2 |
| R1 | FastEthernet0/0 | 192.168.10.10 | 255.255.255.0 | - | LAN Interface |
| R1 | FastEthernet1/0 | 10.10.10.10 | 255.255.255.0 | - | Server Network Interface |
| Server | FastEthernet0 | 10.10.10.11 | 255.255.255.0 | 10.10.10.10 | Server |

**Procedure**

1. Open Cisco Packet Tracer and place one router, one switch, three PCs and one server in the workspace.
2. Connect PC0, PC1 and PC2 to the switch using Copper Straight-Through cables.
3. Connect the switch to the router using a Copper Straight-Through cable.
4. Connect the router to the server using a Copper Straight-Through cable.
5. Configure the IP address of PC0:
   - IP Address: 192.168.10.1
   - Default Gateway: 192.168.10.10
6. Configure the IP address of PC1:
   - IP Address: 192.168.10.2
   - Default Gateway: 192.168.10.10
7. Configure the IP address of PC2:
   - IP Address: 192.168.10.3
   - Default Gateway: 192.168.10.10
8. Configure the Server:
   - IP Address: 10.10.10.11
   - DNS Server: 0.0.0.0

   The above PC and server configuration follows the values in the uploaded ACL experiment.
9. Configure the router interfaces:

   Router → Config → FastEthernet 0/0

   IP Address → 192.168.10.10

   Router → Config → FastEthernet 1/0

   IP Address → 10.10.10.10
10. Select Router → CLI.
11. Enter privileged EXEC mode:

    Router> enable
12. Enter global configuration mode:

    Router# configure terminal
13. Create a Standard Access Control List numbered 11:

    Router(config)# ip access-list standard 11

    Router(config-std-nacl)# deny host 192.168.10.2

    Router(config-std-nacl)# permit any

Router(config-std-nacl)# exit

14. Apply ACL 11 to the router interface connected to the user network:

    Router(config)# interface fa0/0

    Router(config-if)# ip access-group 11 in

    Router(config-if)# exit

    The uploaded manual uses the same ACL number, denied host, permit any, and inbound application on Fa0/0.

15. Save the configuration:

    Router(config)# end

    Router# copy running-config startup-config

16. Test connectivity from PC0 to the server:

    PC> ping 10.10.10.11

17. Test connectivity from PC1 to the server:

    PC> ping 10.10.10.11

18. Test connectivity from PC2 to the server:

    PC> ping 10.10.10.11

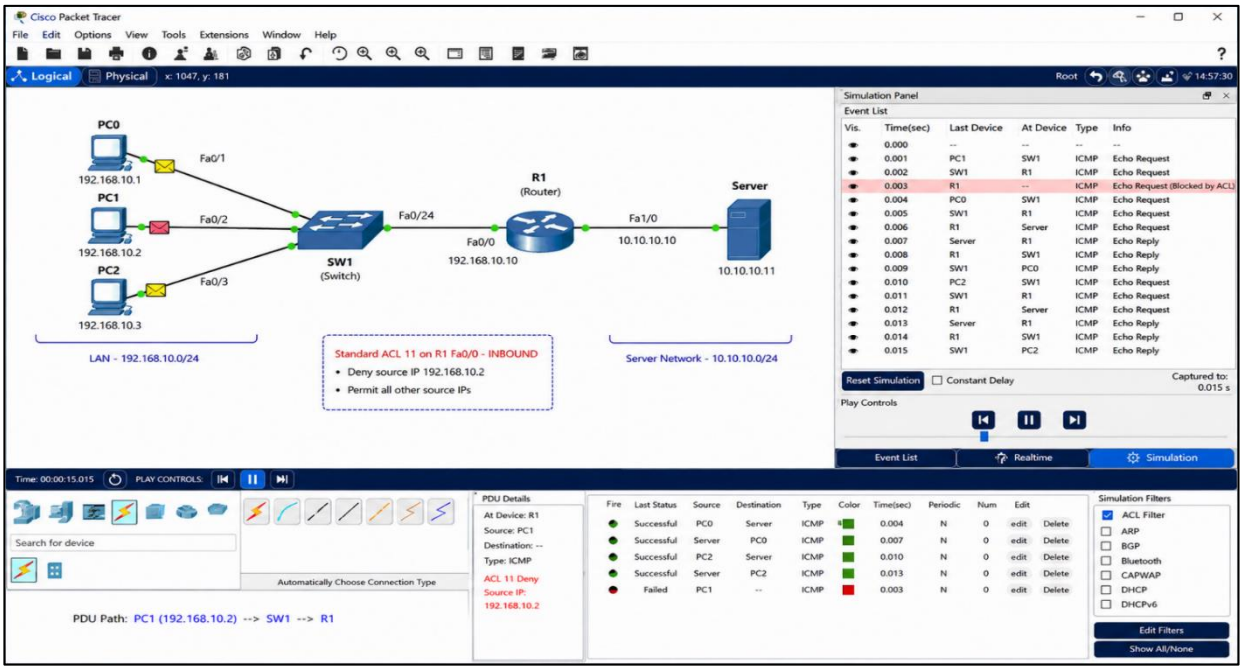
19. Verify that access from PC1 (192.168.10.2) is denied while access from the other PCs is permitted.

20. Verify the ACL configuration using:

    Router# show access-lists

21. Verify the interface configuration using:

    Router# show ip interface brief



**9. Configuration of Extended Access Control Lists (ACLs) and Analysis using Wireshark**

**In enterprise networks, controlling traffic based on multiple parameters such as IP address, protocol, and port number is essential for security. This is achieved using Extended Access Control List. To verify and analyze how ACL rules affect traffic, the administrator uses Wireshark**

**Aim**: - To configure an Extended Access Control List on a Cisco router based on source IP address, destination IP address, protocol, and port number, and analyze the effect of ACL rules on network traffic using Wireshark.

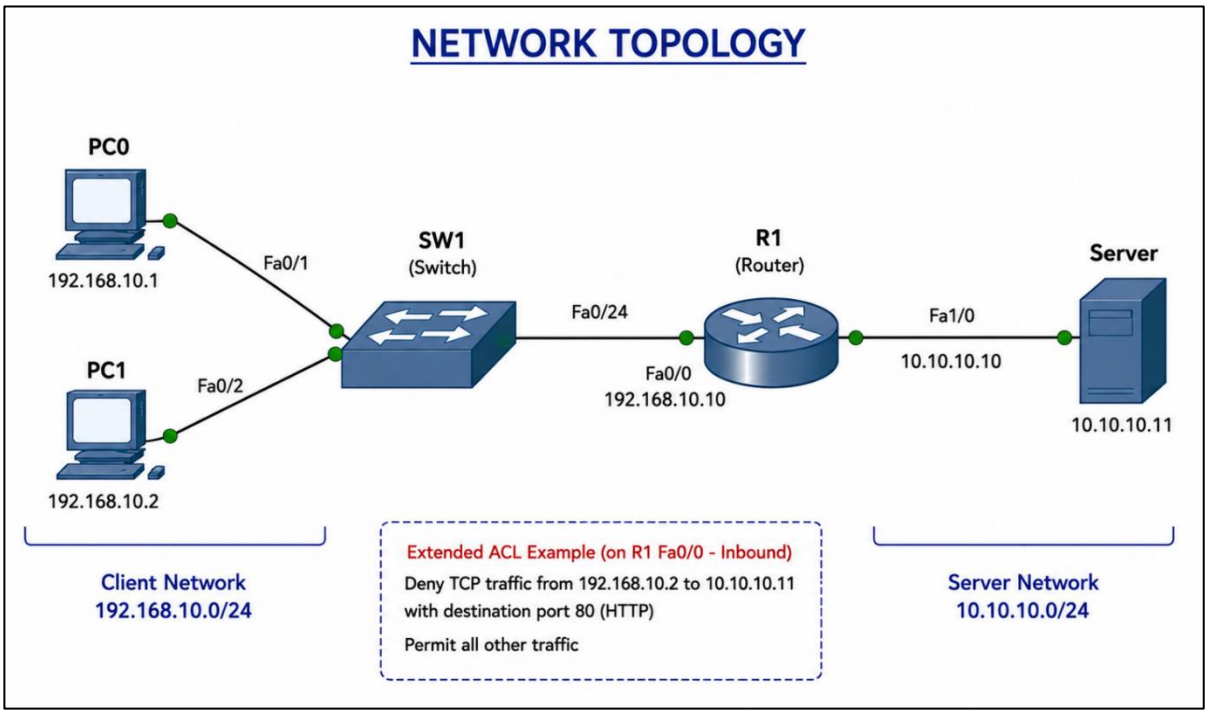
**Apparatus Required**

1. Cisco Packet Tracer
2. Cisco Router
3. Cisco Switch
4. PCs (2)
5. Server
6. Wireshark
7. Copper Straight-Through Cables

**Network Topology**

The network consists of two PCs connected to a switch. The switch is connected to Router R1, which is connected to a server network.

- PC0 → 192.168.10.1
- PC1 → 192.168.10.2
- R1 Fa0/0 → 192.168.10.10
- R1 Fa1/0 → 10.10.10.10
- Server → 10.10.10.11




## IP ADDRESS CONFIGURATION

| Device | Interface | IP Address | Subnet Mask | Default Gateway | Description |
|---|---|---|---|---|---|
| PC0 | FastEthernet0 | 192.168.10.1 | 255.255.255.0 | 192.168.10.10 | Client PC 0 |
| PC1 | FastEthernet0 | 192.168.10.2 | 255.255.255.0 | 192.168.10.10 | Client PC 1 (Traffic restricted by ACL) |
| R1 | FastEthernet0/0 | 192.168.10.10 | 255.255.255.0 | - | Interface towards Client Network |
| R1 | FastEthernet1/0 | 10.10.10.10 | 255.255.255.0 | - | Interface towards Server Network |
| Server | FastEthernet0 | 10.10.10.11 | 255.255.255.0 | 10.10.10.10 | Web Server |

**Procedure**

1. Open Cisco Packet Tracer and place one router, one switch, two PCs, and one server in the workspace.
2. Connect PC0 and PC1 to the switch using Copper Straight-Through cables.
3. Connect the switch to Router R1.
4. Connect Router R1 to the server.
5. Configure the IP address of PC0:

- IP Address: 192.168.10.1
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.10

6. Configure the IP address of PC1:
   - IP Address: 192.168.10.2
   - Subnet Mask: 255.255.255.0
   - Default Gateway: 192.168.10.10
7. Configure Router R1:
   - Fa0/0 IP Address: 192.168.10.10
   - Fa1/0 IP Address: 10.10.10.10
8. Configure the Server:
   - IP Address: 10.10.10.11
   - Subnet Mask: 255.255.255.0
   - Default Gateway: 10.10.10.10
9. Select Router → CLI.
10. Enter privileged EXEC mode:

    Router> enable
11. Enter global configuration mode:

    Router# configure terminal
12. Create an Extended ACL:

    Router(config)# ip access-list extended 101
13. Deny HTTP traffic from PC1 to the server:

    Router(config-ext-nacl)# deny tcp host 192.168.10.2 host 10.10.10.11 eq 80
14. Permit all other IP traffic:

    Router(config-ext-nacl)# permit ip any any

    Router(config-ext-nacl)# exit
15. Apply the ACL to the appropriate router interface in the required direction:

    Router(config)# interface fa0/0

    Router(config-if)# ip access-group 101 in

    Router(config-if)# exit
16. Save the configuration:

    Router(config)# end

    Router# copy running-config startup-config

17. Verify the ACL:

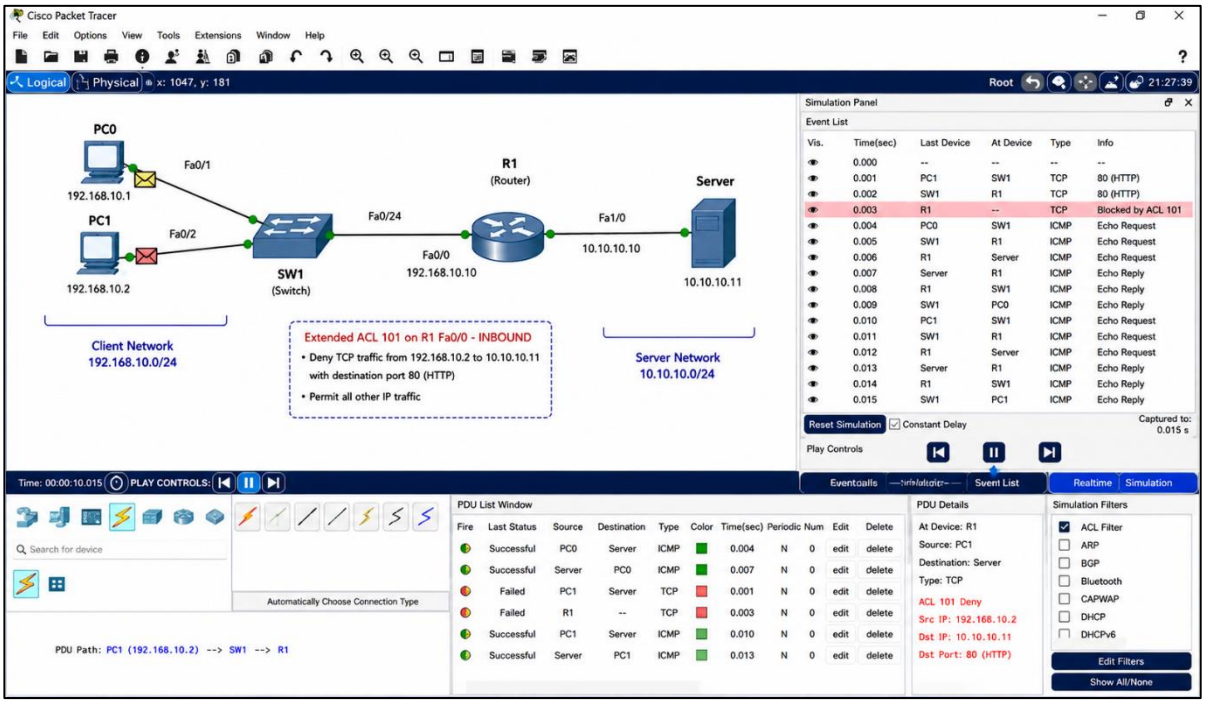
    Router# show access-lists

18. From PC1, test connectivity to the server.

    PC> ping 10.10.10.11

19. Test the HTTP service from PC1 and observe whether the traffic is permitted or denied according to the ACL rule.
20. Test communication from PC0 to verify that traffic not matching the denied rule is permitted.
21. Open Wireshark on the system/interface through which the traffic is being captured.
22. Start packet capture and generate traffic between the client and server.
23. Observe the captured packets and examine fields such as:
    - Source IP address
    - Destination IP address
    - Protocol
    - Source port
    - Destination port
    - TCP/UDP information
24. Apply suitable Wireshark display filters to analyze the generated traffic.
25. Compare the captured traffic before and after applying the Extended ACL and observe the difference in permitted and denied traffic.
26. Stop the packet capture and analyze the packets to verify the effect of the ACL.

**Output**



**10. Implementation of Network Address Resolution using Wireshark.**

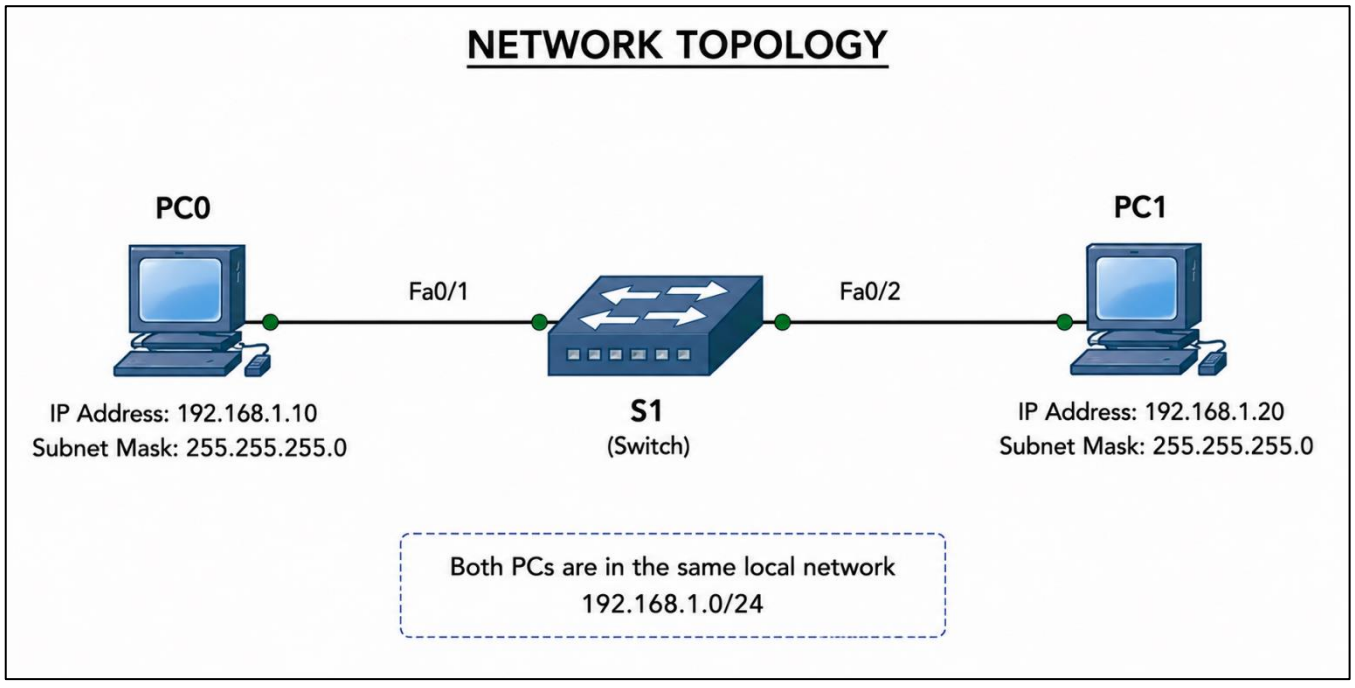
**A system in a local network wants to communicate with another device using its IP address. However, to send data at the data link layer, it must first know the destination device's MAC address. The administrator captures network traffic using Wireshark to observe how ARP resolves this issue.**

**Aim: -** To observe and analyze the Address Resolution Protocol (ARP) process using Wireshark and understand how an IP address is resolved to a MAC address for communication within a local network.

**Apparatus Required**

1. Wireshark
2. PC (2)
3. Network Switch
4. Ethernet Cable

**NETWORK TOPOLOGY**



## IP ADDRESS CONFIGURATION

| Device | Interface | IP Address | Subnet Mask | Description |
|---|---|---|---|---|
| **PC0** | FastEthernet0 | 192.168.1.10 | 255.255.255.0 | Source PC |
| **PC1** | FastEthernet0 | 192.168.1.20 | 255.255.255.0 | Destination PC |
| **Switch S1** | - | - | - | Layer 2 Switch |

**Procedure**

1. Open the network environment and connect PC0 and PC1 to Switch S1 using Ethernet cables.
2. Configure the IP address of PC0:
    - IP Address: 192.168.1.10
    - Subnet Mask: 255.255.255.0
3. Configure the IP address of PC1:
    - IP Address: 192.168.1.20
    - Subnet Mask: 255.255.255.0
4. Verify that both PCs belong to the same local network.
5. Open Command Prompt on PC0.
6. Clear the existing ARP cache before starting the experiment.
7. Start Wireshark and select the appropriate network interface for packet capture.
8. Start the packet capture.
9. From PC0, send a ping to PC1:

   ping 192.168.1.20
10. Observe the captured packets in Wireshark.
11. Apply the following display filter to identify ARP packets:

    arp
12. Locate the ARP Request packet.
13. Examine the ARP Request and observe:
    - Source MAC Address

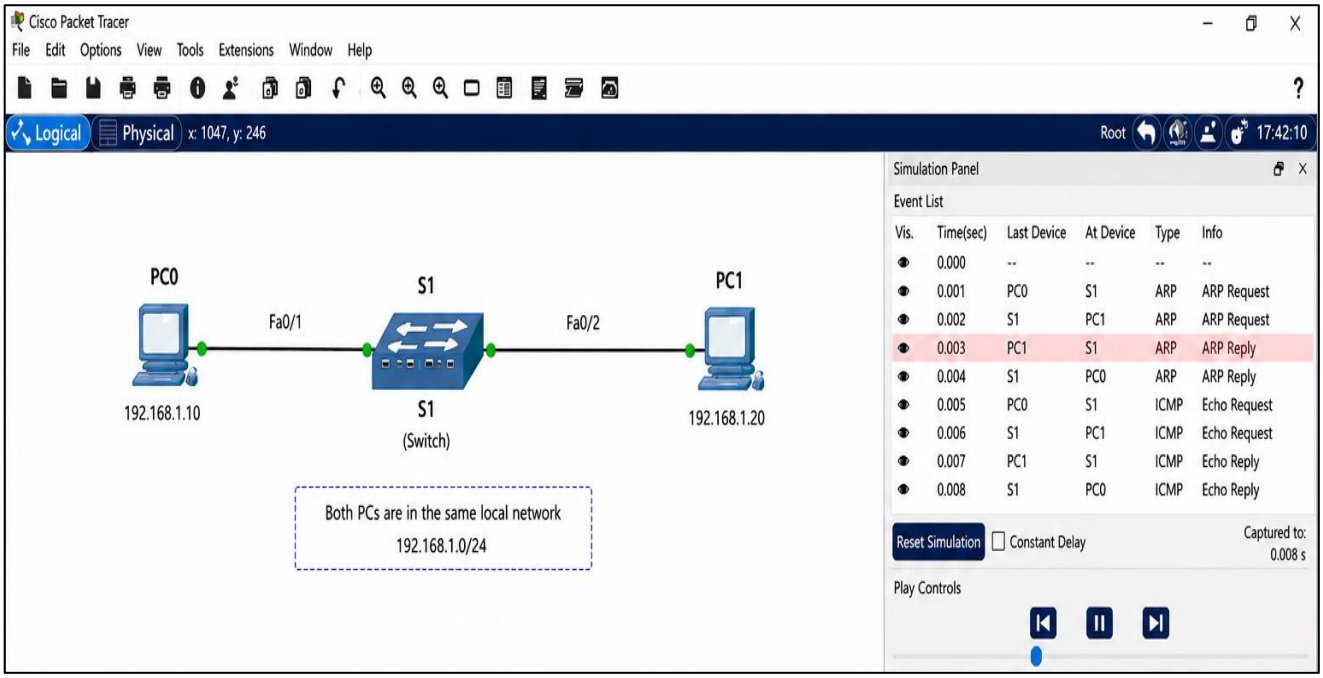
Source IP Address

- Destination IP Address
- Broadcast destination MAC address

14. Locate the corresponding ARP Reply packet.
15. Examine the ARP Reply and observe the MAC address associated with 192.168.1.20.
16. Verify that the ARP exchange occurs before the ICMP Echo Request and Echo Reply packets.
17. Stop the Wireshark capture.
18. Analyze the captured packets and identify the sequence:

    ARP Request → ARP Reply → ICMP Echo Request → ICMP Echo Reply
19. Open the ARP cache on the system and verify that the MAC address corresponding to 192.168.1.20 has been learned.

**Output**



11. **Switch Configuration and Security Implementation using Cisco Packet Tracer.**

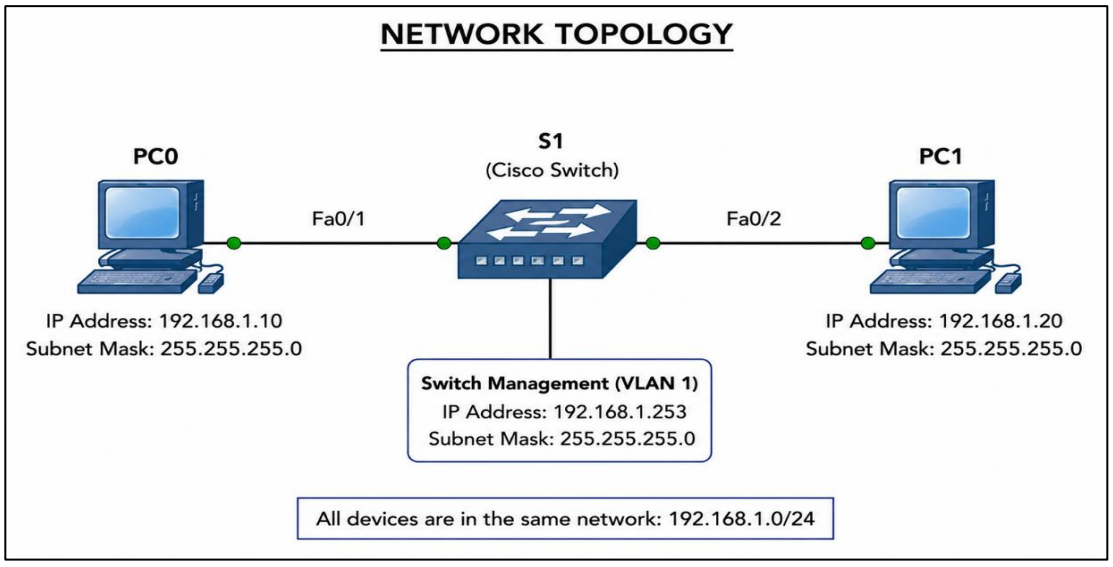
**A college campus network requires secure access to its internal systems. Unauthorized devices connecting to switches can compromise security. To prevent this, the network administrator configures a switch with basic settings, enables port security, and sets up passwords for secure access using Cisco Packet Tracer.**

**Aim**: - To configure a Cisco switch with basic security settings and port security using Cisco Packet Tracer and verify the configuration.

**Apparatus Required**

1. Cisco Packet Tracer
2. Cisco Switch
3. PCs (2)
4. Copper Straight-Through Cables

**Network Topology**



**IP ADDRESS CONFIGURATION TABLE**

| Device | Interface | IP Address | Subnet Mask | Default Gateway | Description |
|---|---|---|---|---|---|
| **PC0** | FastEthernet0 | 192.168.1.10 | 255.255.255.0 | - | Client PC 0 |
| **PC1** | FastEthernet0 | 192.168.1.20 | 255.255.255.0 | - | Client PC 1 |
| **S1 (Switch)** | VLAN 1 (Management) | 192.168.1.253 | 255.255.255.0 | - | Switch Management Interface |

**Procedure**

1. Open Cisco Packet Tracer and place one Cisco switch and two PCs in the workspace.
2. Connect PC0 and PC1 to Switch S1 using Copper Straight-Through cables.
3. Configure the IP address of PC0:
   - IP Address: 192.168.1.10
   - Subnet Mask: 255.255.255.0
4. Configure the IP address of PC1:
   - IP Address: 192.168.1.20
   - Subnet Mask: 255.255.255.0
5. Select Switch S1 → CLI.
6. Enter privileged EXEC mode:
   Switch> enable
7. Enter global configuration mode:
   Switch# configure terminal
8. Configure the hostname:
   Switch(config)# hostname S1
9. Configure the privileged EXEC password:
   S1(config)# enable password cisco
10. Configure the encrypted privileged EXEC password:
    S1(config)# enable secret cisco123
11. Enable password encryption:
    S1(config)# service password-encryption

12. Configure a login warning message:

    S1(config)# banner motd #Authorized Access Only#

13. Configure the console password:

    S1(config)# line console 0

    S1(config-line)# password cisco

    S1(config-line)# login

    S1(config-line)# exit

14. Configure VTY access:

    S1(config)# line vty 0 4

    S1(config-line)# password cisco

    S1(config-line)# login

    S1(config-line)# exit

    These basic security commands follow the security configuration shown in the uploaded manual.

15. Configure the management IP address on VLAN 1:

    S1(config)# interface vlan 1

    S1(config-if)# ip address 192.168.1.253 255.255.255.0

    S1(config-if)# no shutdown

    S1(config-if)# exit

    The reference manual uses 192.168.1.253 / 255.255.255.0 for the switch management interface.

16. Configure port security on the port connected to PC0:

    S1(config)# interface fa0/1

    S1(config-if)# switchport mode access

    S1(config-if)# switchport port-security

    S1(config-if)# switchport port-security maximum 1

    S1(config-if)# switchport port-security mac-address sticky

    S1(config-if)# switchport port-security violation shutdown

    S1(config-if)# exit

17. Configure port security on the port connected to PC1:

    S1(config)# interface fa0/2

    S1(config-if)# switchport mode access

    S1(config-if)# switchport port-security

    S1(config-if)# switchport port-security maximum 1

    S1(config-if)# switchport port-security mac-address sticky

S1(config-if)# switchport port-security violation shutdown

S1(config-if)# exit

18. Save the configuration:

S1(config)# end

S1# copy running-config startup-config

19. Verify the switch configuration:

S1# show running-config

20. Verify the management interface:

S1# show ip interface brief

21. Verify port-security configuration:

S1# show port-security

22. Verify port-security information for individual interfaces:

S1# show port-security interface fa0/1

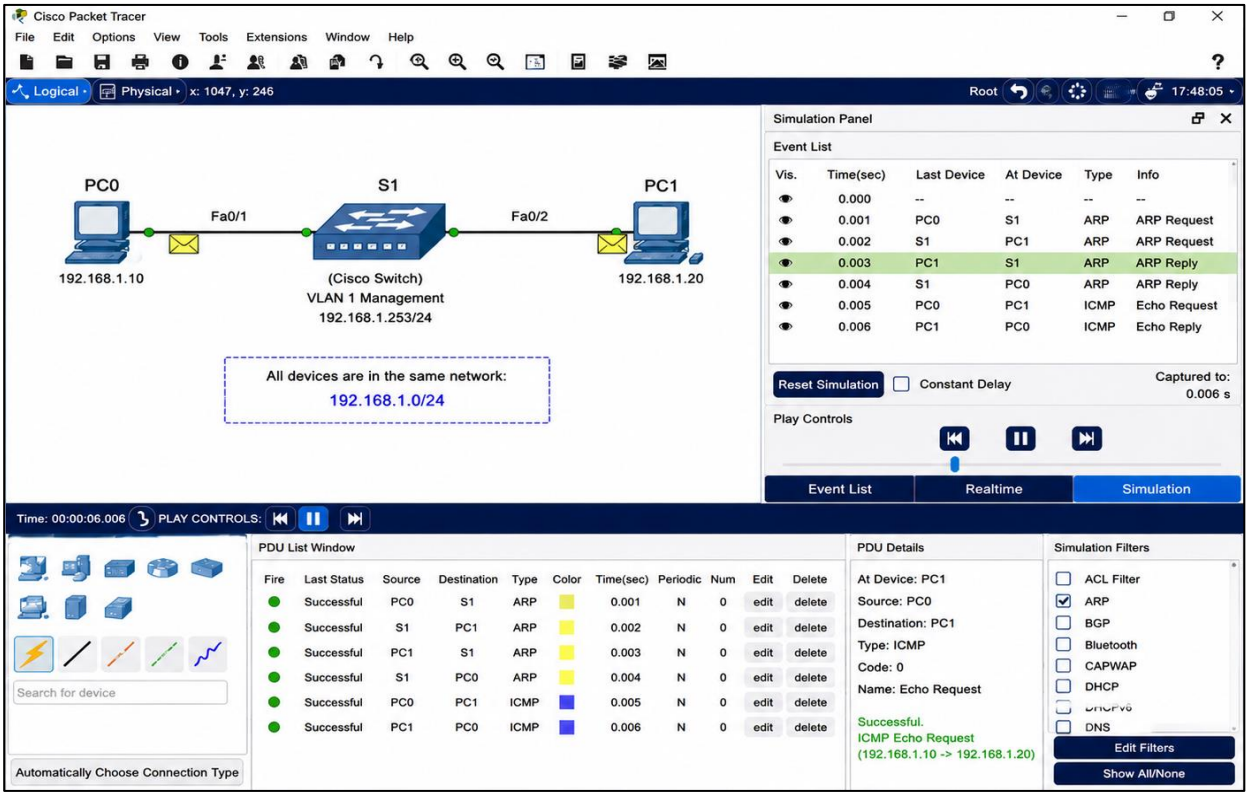
S1# show port-security interface fa0/2

23. From PC0, open Desktop → Command Prompt and test connectivity to PC1:

PC> ping 192.168.1.20

24. Verify that the configured switch ports permit the authorized devices and that an unauthorized MAC address causes the configured port-security violation action.

**Output**



## 12. Router Configuration using Basic Commands in Cisco Packet Tracer.

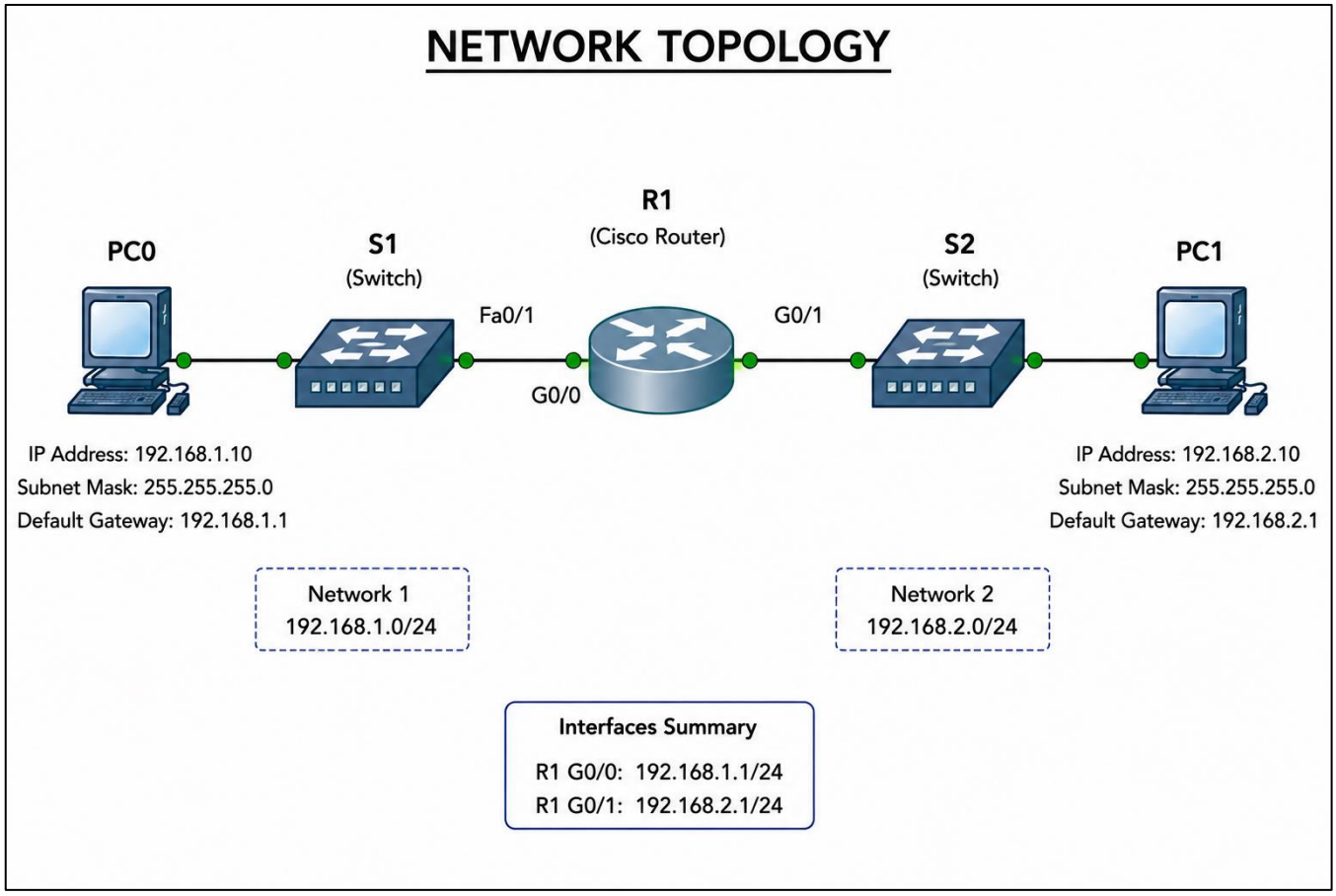
**A small organization is setting up its network to enable communication between different departments and external networks. A router is required to interconnect networks and manage data routing. The network administrator uses Cisco Packet Tracer to configure the router with basic commands and verify connectivity.**

**Aim: -** To configure a Cisco router with basic commands using Cisco Packet Tracer and verify connectivity between different networks.

### Apparatus Required

1. Cisco Packet Tracer
2. Cisco Router
3. Cisco Switches (2)
4. PCs (2)
5. Copper Straight-Through Cables

### Network Topology



## IP ADDRESS CONFIGURATION TABLE

| Device | Interface | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|---|
| **PC0** | FastEthernet0 | 192.168.1.10 | 255.255.255.0 | 192.168.1.1 |
| **PC1** | FastEthernet0 | 192.168.2.10 | 255.255.255.0 | 192.168.2.1 |
| **Router R1** | GigabitEthernet0/0 | 192.168.1.1 | 255.255.255.0 | - |
| **Router R1** | GigabitEthernet0/1 | 192.168.2.1 | 255.255.255.0 | - |

**Procedure**

1. Open Cisco Packet Tracer and place one Cisco router, two switches, and two PCs in the logical workspace.
2. Connect PC0 to Switch S1 using a Copper Straight-Through cable.
3. Connect Switch S1 to Router R1 interface GigabitEthernet0/0.
4. Connect Router R1 interface GigabitEthernet0/1 to Switch S2.
5. Connect Switch S2 to PC1 using a Copper Straight-Through cable.
6. Configure the IP address of PC0:
    - IP Address: 192.168.1.10
    - Subnet Mask: 255.255.255.0
    - Default Gateway: 192.168.1.1
7. Configure the IP address of PC1:
    - IP Address: 192.168.2.10
    - Subnet Mask: 255.255.255.0
    - Default Gateway: 192.168.2.1
8. Configure the router. Select:
   Router R1 → CLI
9. Enter privileged EXEC mode:

Router> enable

10. Enter global configuration mode:

    Router# configure terminal

11. Assign a hostname to the router:

    Router(config)# hostname R1

12. Configure the first router interface:

    R1(config)# interface gigabitEthernet 0/0

    R1(config-if)# ip address 192.168.1.1 255.255.255.0

    R1(config-if)# no shutdown

    R1(config-if)# exit

13. Configure the second router interface:

    R1(config)# interface gigabitEthernet 0/1

    R1(config-if)# ip address 192.168.2.1 255.255.255.0

    R1(config-if)# no shutdown

    R1(config-if)# exit

14. Configure a privileged EXEC password:

    R1(config)# enable secret cisco

15. Configure a login warning message:

    R1(config)# banner motd #Authorized Access Only#

16. Configure the console password:

    R1(config)# line console 0

    R1(config-line)# password cisco

    R1(config-line)# login

    R1(config-line)# exit

17. Configure VTY access:

    R1(config)# line vty 0 4

    R1(config-line)# password cisco

    R1(config-line)# login

    R1(config-line)# exit

18. Save the configuration:

    R1(config)# end

    R1# copy running-config startup-config

19. Verify the configured interfaces using:

R1# show ip interface brief

20. Verify the running configuration using:

    R1# show running-config

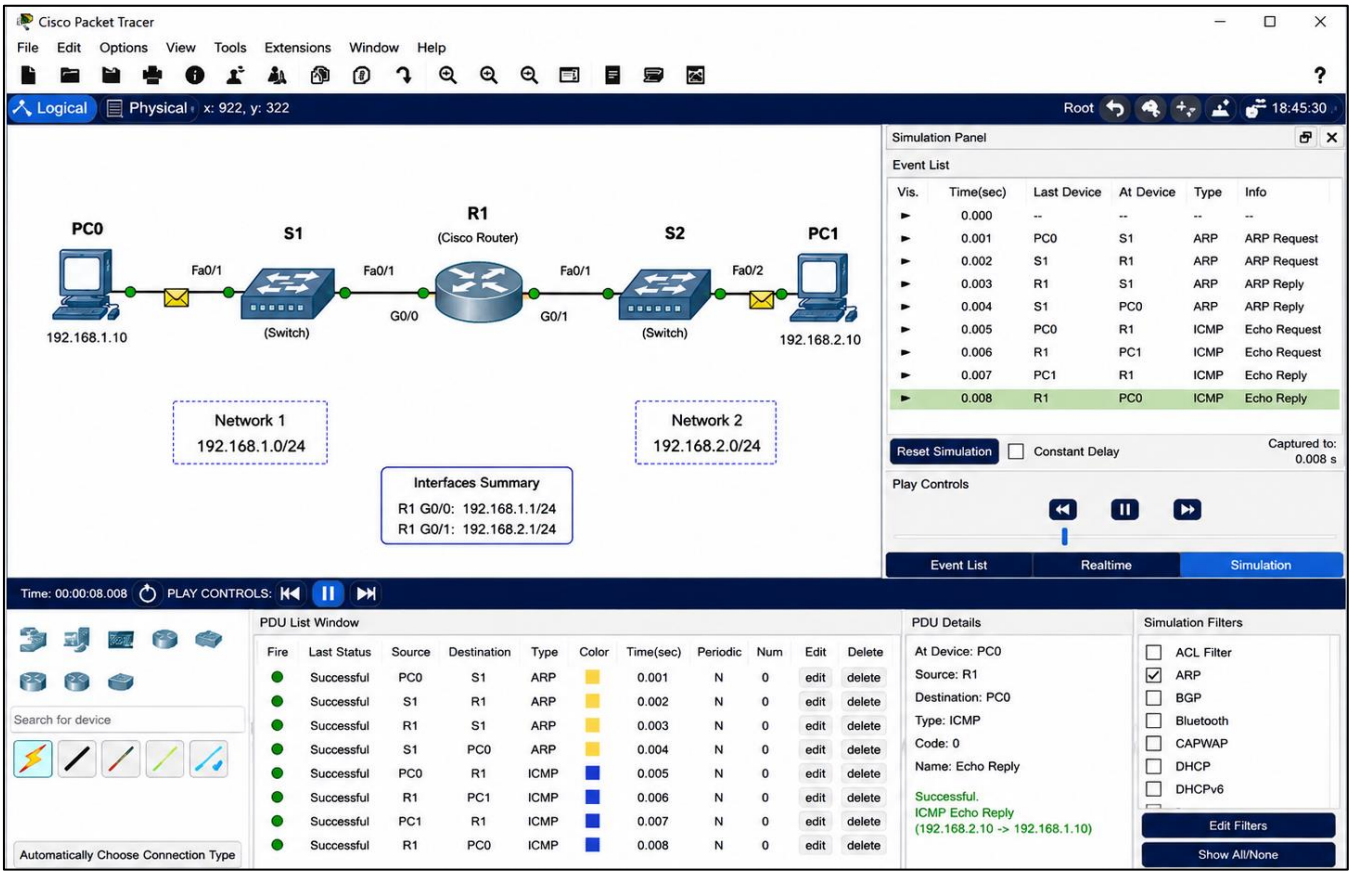
21. From PC0, open Desktop → Command Prompt and test connectivity to PC1 using:

    ping 192.168.2.10

22. If the ping is successful, communication between the two different departmental networks has been established successfully.

**Output**

The router was successfully configured with basic Cisco IOS commands in Cisco Packet Tracer. The IP addresses were assigned to both router interfaces and the interfaces were enabled successfully. The ICMP packet was successfully transmitted from PC0 (192.168.1.10) to PC1 (192.168.2.10) through S1 → R1 → S2, and the return Echo Reply was received successfully. The communication between the two different networks was verified using Simulation Mode.



**13. Switch Configuration and Security Implementation using Cisco Packet Tracer.**

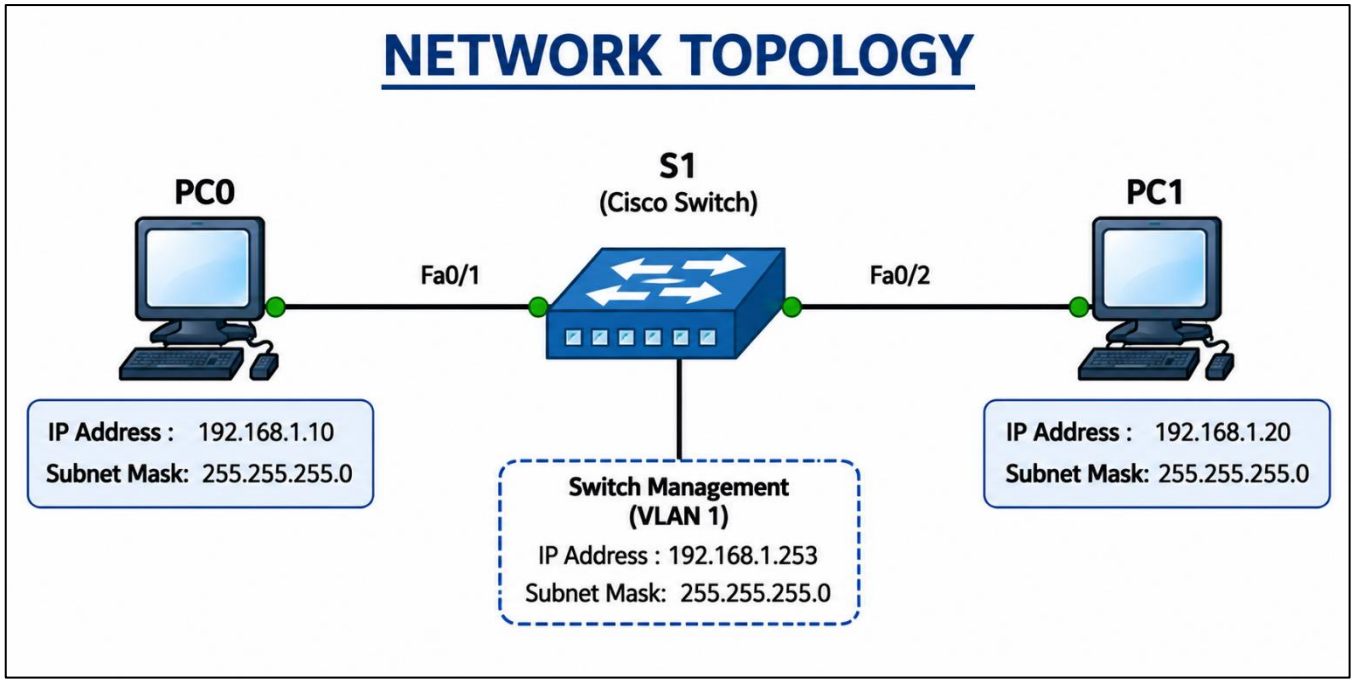
**A college campus network requires secure access to its internal systems. Unauthorized devices connecting to switches can compromise security. To prevent this, the network administrator configures a switch with basic settings, enables port security, and sets up passwords for secure access using Cisco Packet Tracer.**

**Aim**: - To configure a Cisco switch with basic security settings, enable port security, and verify the configuration using Cisco Packet Tracer.

**Apparatus Required**

1. Cisco Packet Tracer
2. Cisco Switch
3. PCs (2)
4. Copper Straight-Through Cables

**Network Topology**



**IP ADDRESSING TABLE**

| Device | Interface | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|---|
| **PC0** | FastEthernet0 | 192.168.1.10 | 255.255.255.0 | - |
| **R1 / S1** | VLAN 1 | 192.168.1.253 | 255.255.255.0 | - |
| **PC1** | FastEthernet0 | 192.168.1.20 | 255.255.255.0 | - |

**Procedure**

1. Open Cisco Packet Tracer and place one Cisco switch and two PCs in the logical workspace.
2. Connect PC0 and PC1 to Switch S1 using Copper Straight-Through cables.
3. Configure the IP address of PC0:
   - IP Address: 192.168.1.10
   - Subnet Mask: 255.255.255.0
4. Configure the IP address of PC1:
   - IP Address: 192.168.1.20
   - Subnet Mask: 255.255.255.0
5. Select Switch S1 → CLI.
6. Enter privileged EXEC mode:
   Switch> enable
7. Enter global configuration mode:
   Switch# configure terminal
8. Configure the hostname:
   Switch(config)# hostname S1
9. Configure the privileged EXEC password:
   S1(config)# enable password cisco
10. Configure the encrypted privileged EXEC password:
    S1(config)# enable secret cisco123

11. Enable password encryption:

    S1(config)# service password-encryption

12. Configure the login warning message:

    S1(config)# banner motd #Authorized Access Only#

13. Configure the console password:

    S1(config)# line console 0

    S1(config-line)# password cisco

    S1(config-line)# login

    S1(config-line)# exit

14. Configure VTY access:

    S1(config)# line vty 0 4

    S1(config-line)# password cisco

    S1(config-line)# login

    S1(config-line)# exit

15. Configure the switch management IP address:

    S1(config)# interface vlan 1

    S1(config-if)# ip address 192.168.1.253 255.255.255.0

    S1(config-if)# no shutdown

    S1(config-if)# exit

    The uploaded manual specifically uses VLAN 1 with management IP 192.168.1.253 and the same basic security configuration sequence.

16. Configure port security on the port connected to PC0:

    S1(config)# interface fa0/1

    S1(config-if)# switchport mode access

    S1(config-if)# switchport port-security

    S1(config-if)# switchport port-security maximum 1

    S1(config-if)# switchport port-security mac-address sticky

    S1(config-if)# switchport port-security violation shutdown

    S1(config-if)# exit

17. Configure port security on the port connected to PC1:

    S1(config)# interface fa0/2

    S1(config-if)# switchport mode access

    S1(config-if)# switchport port-security

S1(config-if)# switchport port-security maximum 1

S1(config-if)# switchport port-security mac-address sticky

S1(config-if)# switchport port-security violation shutdown

S1(config-if)# exit

18. Save the configuration:

    S1(config)# end

    S1# copy running-config startup-config

19. Verify the running configuration:

    S1# show running-config

20. Verify the VLAN 1 management interface:

    S1# show ip interface brief

21. Verify port-security configuration:

    S1# show port-security

22. Verify port security on individual interfaces:

    S1# show port-security interface fa0/1

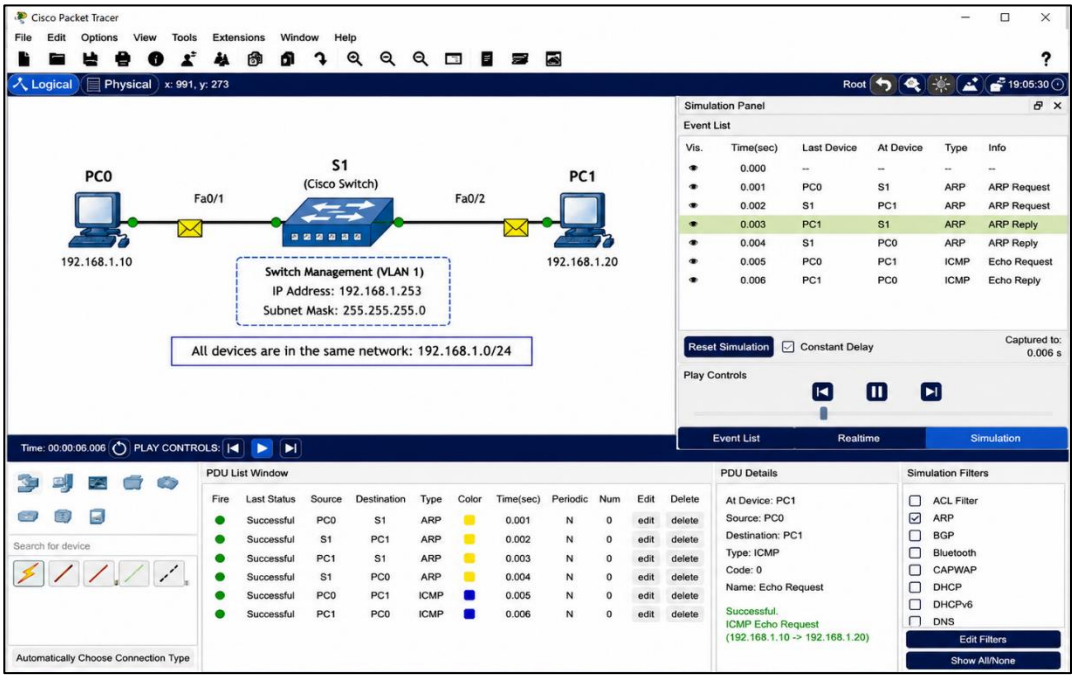
    S1# show port-security interface fa0/2

23. From PC0, open Desktop → Command Prompt and test connectivity to PC1:

    PC> ping 192.168.1.20

24. Verify that the configured ports accept the authorized devices and that an unauthorized MAC address causes the configured port-security violation action.

**Output**



## 14. Router Configuration using Basic Commands in Cisco Packet Tracer.

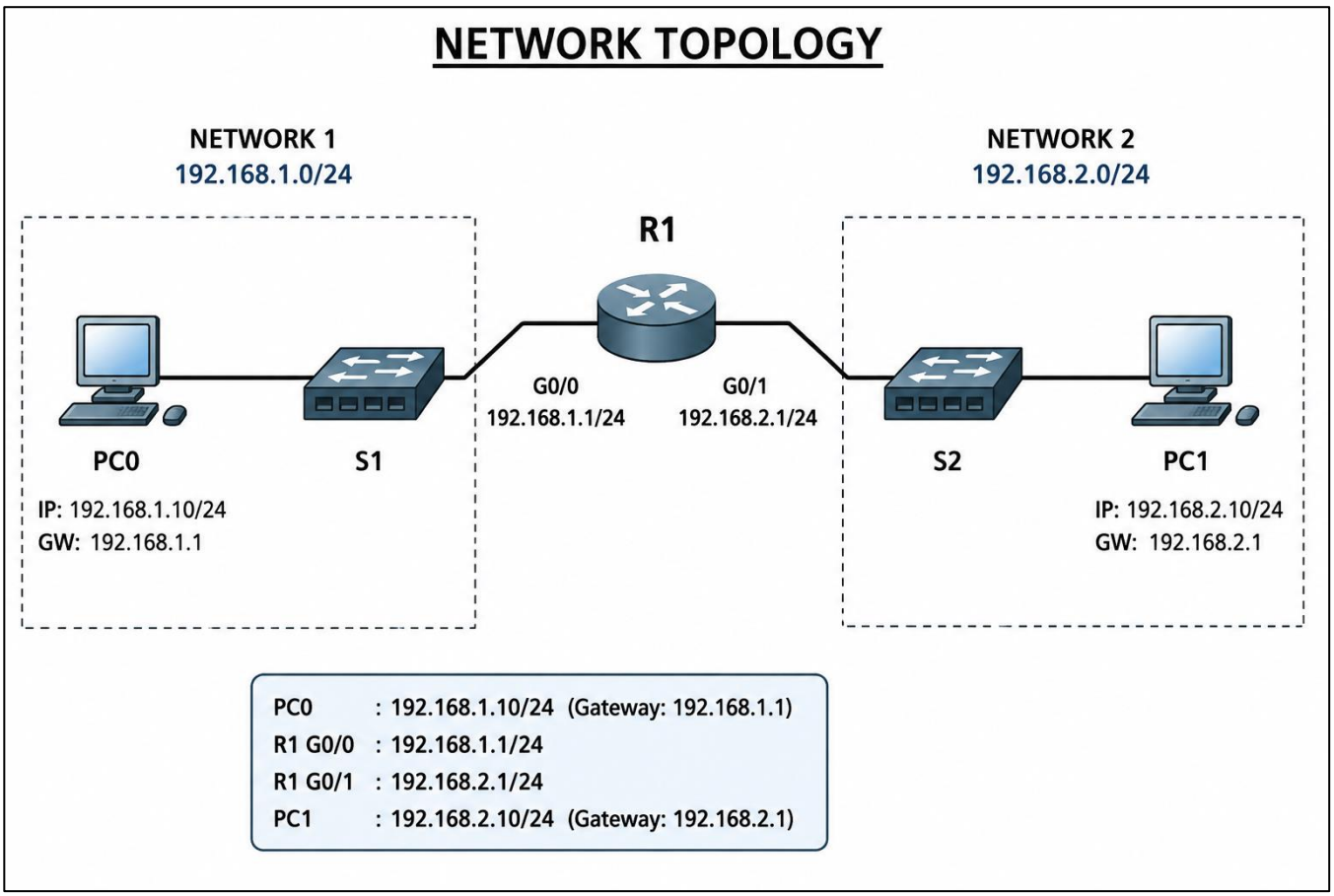
**A small organization is setting up its network to enable communication between different departments and external networks. A router is required to interconnect networks and manage data routing. The network administrator uses Cisco Packet Tracer to configure the router with basic commands and verify connectivity.**

**Aim**: - To configure a Cisco router with basic commands using Cisco Packet Tracer and verify connectivity between different networks.

### Apparatus Required

1. Cisco Packet Tracer
2. Cisco Router
3. Cisco Switches (2)
4. PCs (2)
5. Copper Straight-Through Cables

### Network Topology



## IP ADDRESSING TABLE

| Device | Interface | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|---|
| PC0 | FastEthernet0 | 192.168.1.10 | 255.255.255.0 | 192.168.1.1 |
| R1 | GigabitEthernet0/0 | 192.168.1.1 | 255.255.255.0 | - |
| R1 | GigabitEthernet0/1 | 192.168.2.1 | 255.255.255.0 | - |
| PC1 | FastEthernet0 | 192.168.2.10 | 255.255.255.0 | 192.168.2.1 |

**Procedure**

1. Open Cisco Packet Tracer and place one Cisco router, two switches, and two PCs in the logical workspace.
2. Connect PC0 to Switch S1 using a Copper Straight-Through cable.
3. Connect Switch S1 to Router R1 interface GigabitEthernet0/0.
4. Connect Router R1 interface GigabitEthernet0/1 to Switch S2.
5. Connect Switch S2 to PC1 using a Copper Straight-Through cable.
6. Configure the IP address of PC0:
    - IP Address: 192.168.1.10
    - Subnet Mask: 255.255.255.0
    - Default Gateway: 192.168.1.1
7. Configure the IP address of PC1:
    - IP Address: 192.168.2.10
    - Subnet Mask: 255.255.255.0
    - Default Gateway: 192.168.2.1
8. Select Router R1 → CLI.
9. Enter privileged EXEC mode:

   Router> enable

10. Enter global configuration mode:

    Router# configure terminal

11. Configure the hostname:

    Router(config)# hostname R1

12. Configure the first router interface:

    R1(config)# interface gigabitEthernet 0/0

    R1(config-if)# ip address 192.168.1.1 255.255.255.0

    R1(config-if)# no shutdown

    R1(config-if)# exit

13. Configure the second router interface:

    R1(config)# interface gigabitEthernet 0/1

    R1(config-if)# ip address 192.168.2.1 255.255.255.0

    R1(config-if)# no shutdown

    R1(config-if)# exit

14. Configure the privileged EXEC password:

    R1(config)# enable secret cisco

15. Configure the login warning message:

    R1(config)# banner motd #Authorized Access Only#

16. Configure the console password:

    R1(config)# line console 0

    R1(config-line)# password cisco

    R1(config-line)# login

    R1(config-line)# exit

17. Configure VTY access:

    R1(config)# line vty 0 4

    R1(config-line)# password cisco

    R1(config-line)# login

    R1(config-line)# exit

18. Save the configuration:

    R1(config)# end

    R1# copy running-config startup-config

19. Verify the configured interfaces:

    R1# show ip interface brief

20. Verify the running configuration:

    R1# show running-config

21. From PC0, open Desktop → Command Prompt.

22. Test connectivity to PC1:

    PC> ping 192.168.2.10

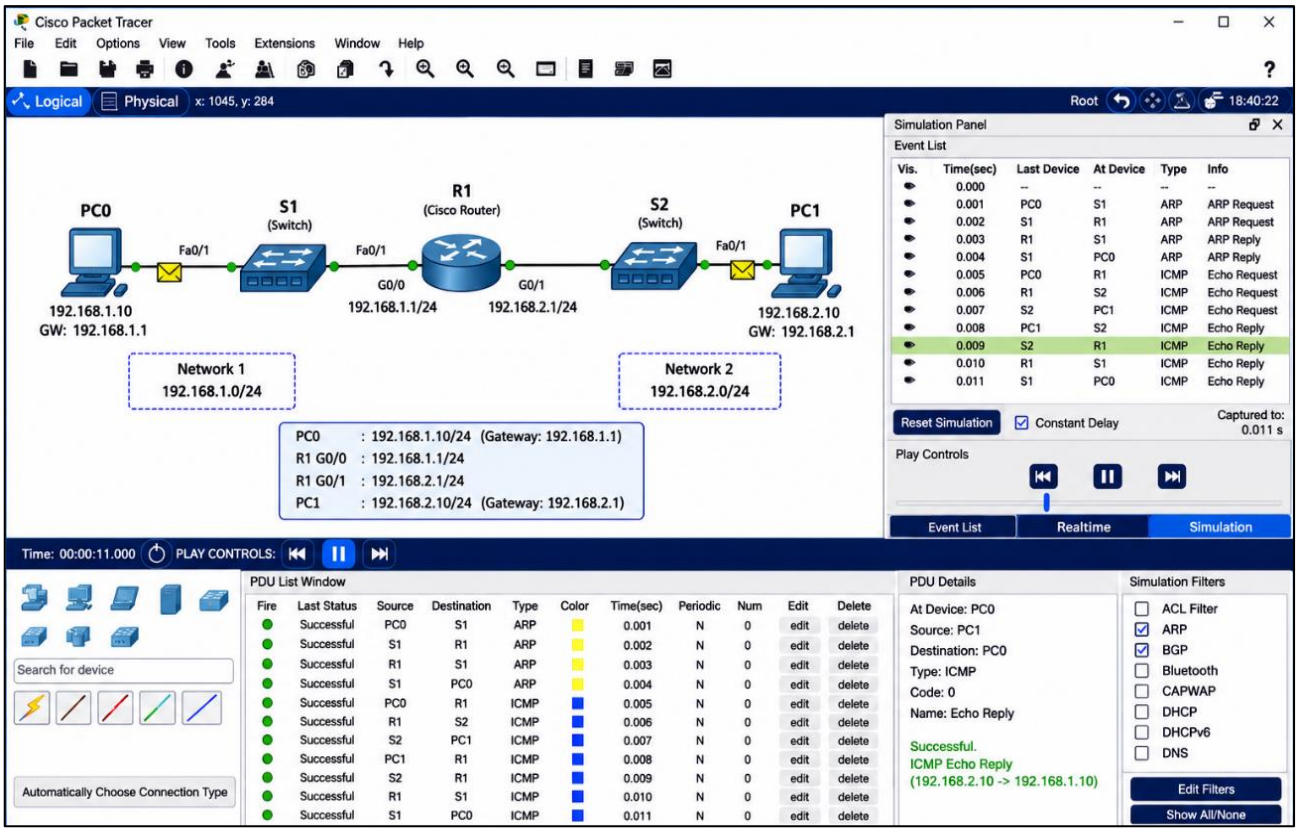
23. Verify that the ping is successful and that communication between the two networks is established.

**Output**

Here is the message simulation for Program 14. The packet travels:

PC0 (192.168.1.10) → S1 → R1 → S2 → PC1 (192.168.2.10)

The yellow envelopes show the ARP process, followed by the blue ICMP Echo Request/Reply packets, with the Simulation Panel and PDU list showing successful delivery.



## VIVA QUESTIONS

1. **What is a computer network?**

   A computer network is a collection of interconnected devices that communicate and share resources.

2. **What is an IP address?**

   An IP address is a logical address used to identify a device on a network.

3. **What is a MAC address?**

   A MAC address is the hardware address assigned to a network interface.

4. **What is the difference between a switch and a router?**

   A switch connects devices within a LAN, while a router connects different networks.

5. **What is a subnet mask?**

   A subnet mask identifies the network portion and host portion of an IP address.

6. **What is a default gateway?**

   A default gateway is the router interface used to reach devices outside the local network.

7. **What is Cisco Packet Tracer?**

   Cisco Packet Tracer is a network simulation tool used to design, configure, and test networks.

8. **What is Simulation Mode in Cisco Packet Tracer?**

   Simulation Mode allows users to observe packets as they travel through the network.

9. **What is the purpose of the enable command?**

   It enters privileged EXEC mode on a Cisco device.

10. **What is the purpose of configure terminal?**

    It enters global configuration mode.

11. **What is the purpose of the no shutdown command?**

    It activates a network interface.

12. **What does show ip interface brief display?**

    It displays interface names, IP addresses, and interface/status information.

13. **What is static routing?**

    Static routing is a method where routes are manually configured by the network administrator.

14. **What is a default route?**

    A default route is used when no specific route exists for a destination network.

15. **What is dynamic routing?**

    Dynamic routing allows routers to automatically exchange routing information and learn routes.

16. **What is RIP?**

RIP is a dynamic routing protocol that uses hop count as its routing metric.

17. **What is the maximum hop count in RIP?**

    The maximum usable hop count is 15.

18. **What is EIGRP?**

    EIGRP is a dynamic routing protocol used for efficient and fast routing between networks.

19. **What is the purpose of no auto-summary in EIGRP?**

    It disables automatic classful route summarization.

20. **What is a VLAN?**

    A VLAN is a logical network segment that creates a separate broadcast domain.

21. **Why are VLANs used?**

    VLANs are used to divide a network into logical segments for better organization, security, and traffic management.

22. **What is Inter-VLAN Routing?**

    Inter-VLAN Routing allows communication between devices in different VLANs.

23. **What is a trunk port?**

    A trunk port carries traffic belonging to multiple VLANs between network devices.

24. **What is VTP?**

    VTP stands for VLAN Trunking Protocol and is used to distribute VLAN information between switches.

25. **What are the VTP modes?**

    The main VTP modes are Server, Client, and Transparent.

26. **What is an ACL?**

    An Access Control List is a set of rules used to permit or deny network traffic.

27. **What is a Standard ACL?**

    A Standard ACL filters traffic primarily based on the source IP address.

28. **What is an Extended ACL?**

    An Extended ACL can filter traffic based on source IP, destination IP, protocol, and port number.

29. **What is the difference between Standard and Extended ACLs?**

    Standard ACLs mainly filter using source IP addresses, while Extended ACLs provide more specific filtering using source, destination, protocol, and ports.

30. **What is port security?**

    Port security restricts which devices or MAC addresses can access a switch port.

31. **What is sticky MAC address learning?**

Sticky MAC allows the switch to dynamically learn a connected device's MAC address and use it as a secure MAC address.

32. **What does violation shutdown do?**

It shuts down the switch port when a port-security violation occurs.

33. **What is ARP?**

RP stands for Address Resolution Protocol and is used to map an IPv4 address to a MAC address.

34. **What is an ARP Request?**

An ARP Request is a broadcast message asking which device owns a particular IP address.

35. **What is an ARP Reply?**

An ARP Reply provides the MAC address associated with the requested IP address.

36. **What is Wireshark?**

Wireshark is a network protocol analyzer used to capture and inspect network packets.

37. **What is the purpose of the ping command?**

It is used to test connectivity and reachability between network devices.

38. **What does show ip route display?**

It displays the routing table and the routes known to the router.

39. **What does show access-lists display?**

It displays the configured ACL rules and their activity.

40. **What does show port-security display?**

It displays the port-security configuration and secure MAC address information.